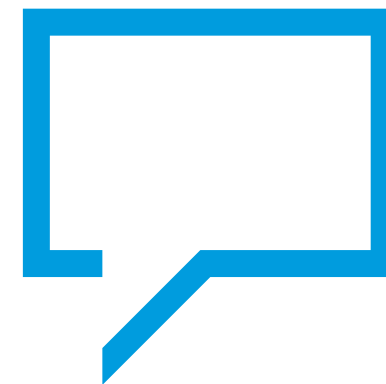




AI IN HEALTHCARE

CMED 301

COLLEGE OF MEDICINE, KING SAUD UNIVERSITY

Shauna M Overgaard, PhD

Center for Digital Health
Mayo Clinic

DISCLOSURES

- I do not intend to discuss an off-label/investigative use of a commercial product/device.
- I do not have financial relationships with entities producing, marketing, selling, reselling, or distributing health care products used by or on patients.

LEARNING OBJECTIVES

1. Define the Role of AI in Healthcare
2. Identify Key AI Concepts
3. Recognize AI Applications in Diagnosis and Treatment
4. Discuss Enhanced Patient Care with AI
5. Analyze Ethical and Regulatory Aspects
6. Anticipate Future AI Trends in Healthcare
7. Analyze Case Studies and Real-World Examples



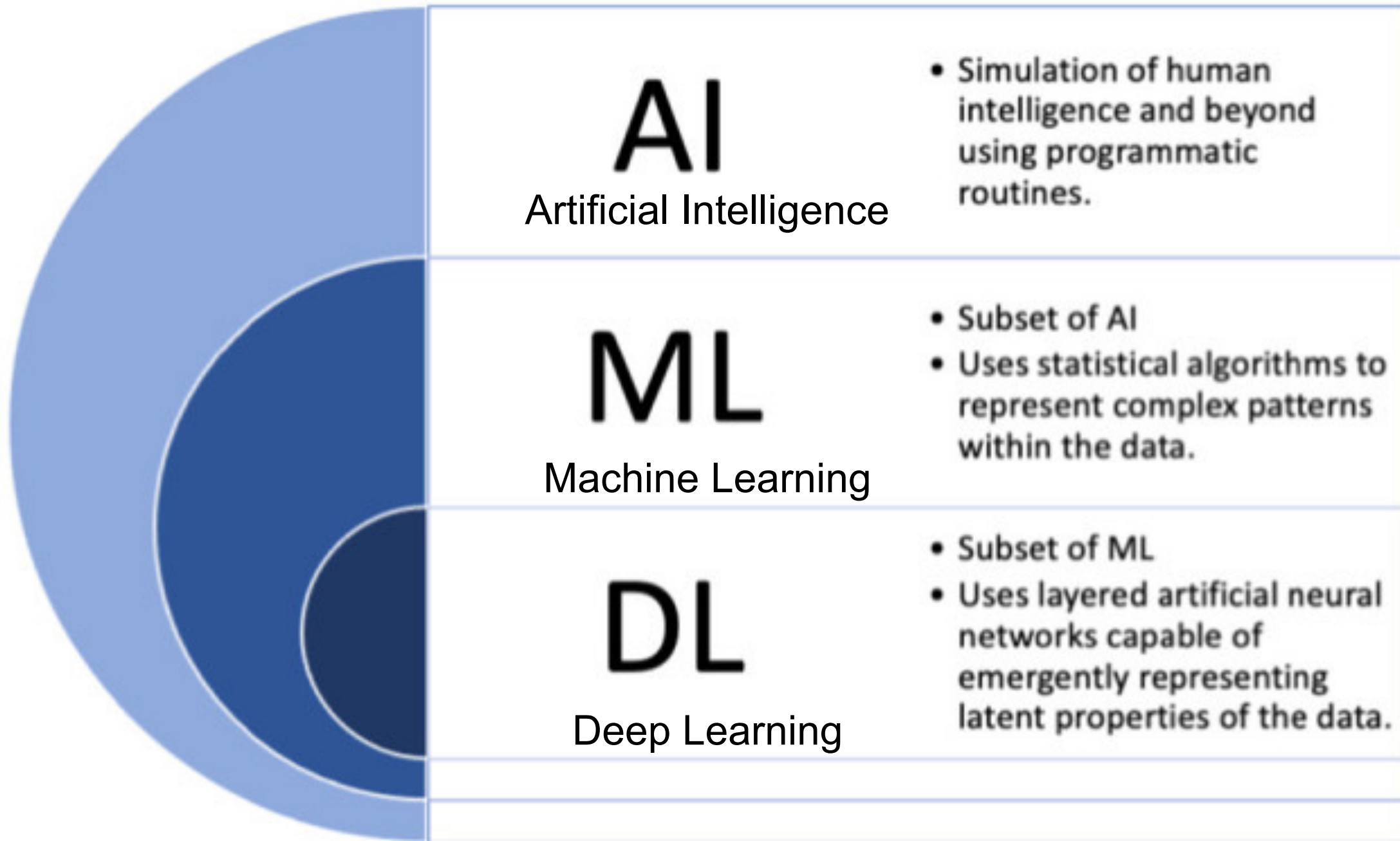
1

DEFINE THE ROLE OF AI IN HEALTHCARE

WHAT IS AI?

artificial intelligence

computer systems that recognize patterns



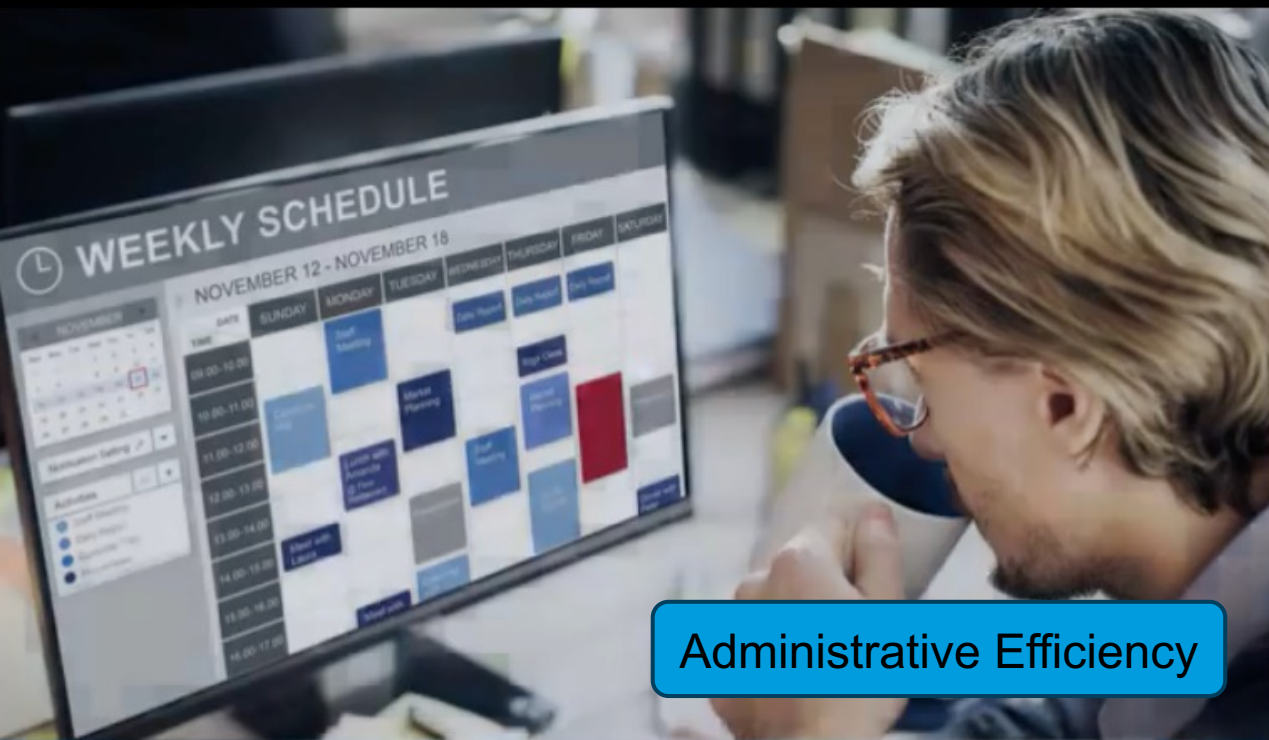
Why use AI in healthcare?

improve patient outcomes

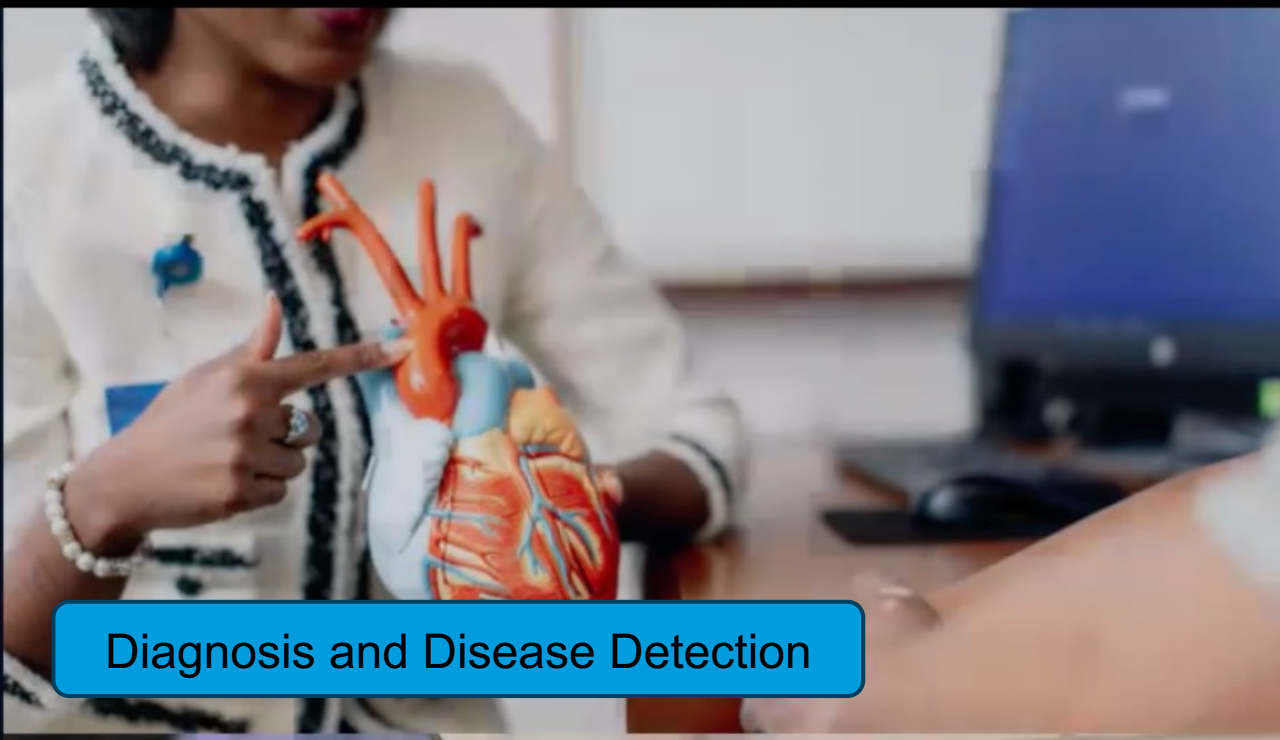
automate processes and save time

reduce costs





Administrative Efficiency



Diagnosis and Disease Detection



Predictive Analytics



Drug and Discovery Development



**Creativity
Empathy
Compassion**



2

IDENTIFY KEY AI CONCEPTS



Machine learning is not magic

- It learns patterns; it does not explain causation
- It cannot replace clinical experience, domain knowledge, and clinical decisions
- It cannot replace clinical intuition and qualitative assessment
- It cannot handle rare, unpredictable events

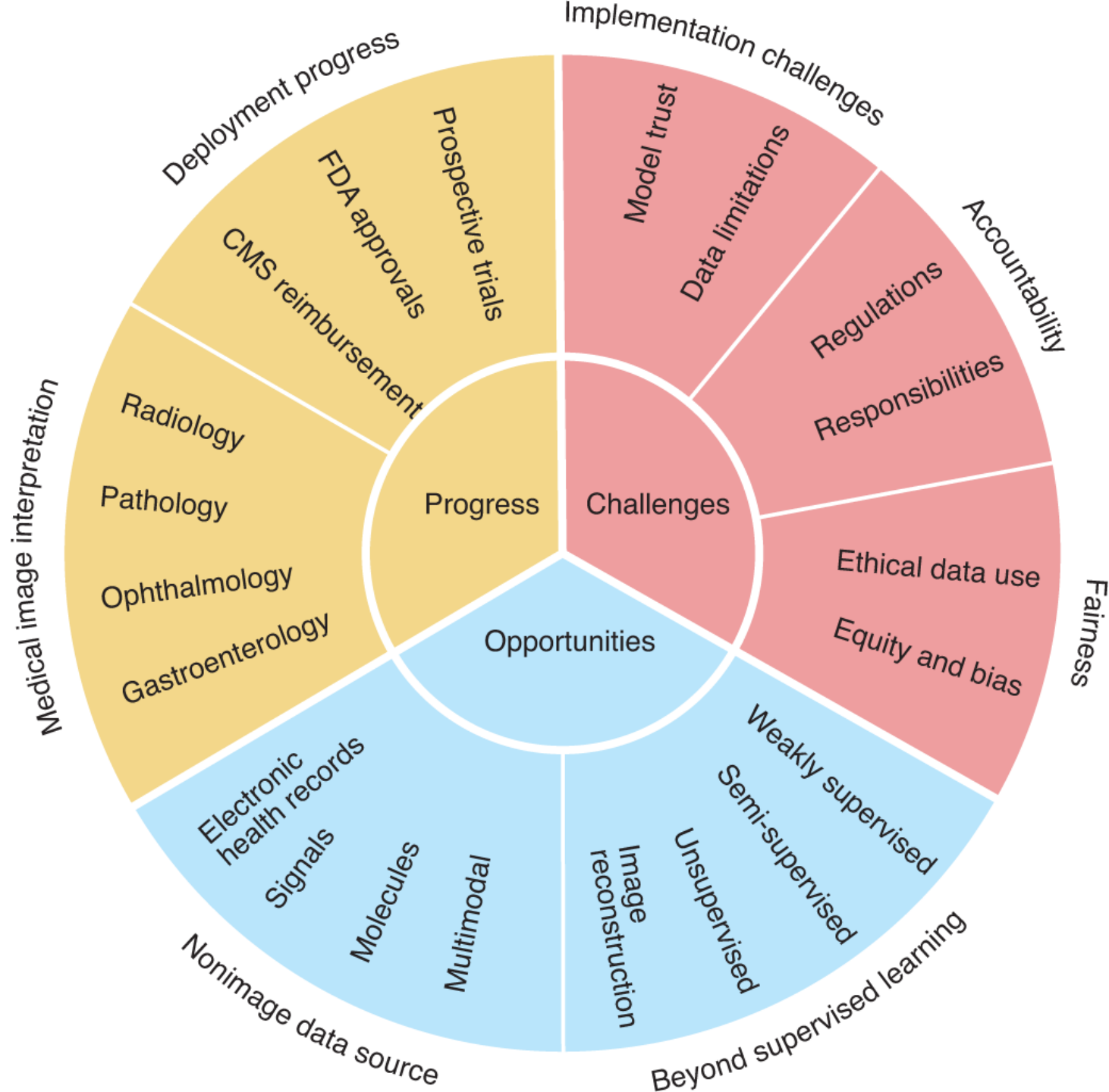
Credit: Joshua Reed, PhD

ARTIFICIAL INTELLIGENCE KEY CONCEPTS

Self-supervised learning	Learning from unlabeled data by leveraging information extracted from the data itself
Semi-supervised learning	Learning from a small amount of labeled data combined with a large amount of unlabeled data
Causal inference	Finding the effect of a component or treatment on a system using data
Reinforcement learning	Learning in an interactive environment using feedback from actions and past experiences

AI setups beyond supervised learning

OVERVIEW OF THE PROGRESS, CHALLENGES AND OPPORTUNITIES FOR AI IN HEALTH





3

RECOGNIZE AI APPLICATIONS IN DIAGNOSIS AND TREATMENT

Examples of applications of AI in healthcare

radiology	sepsis	readmission	pathology	face recognition
pharmacy	surgery	risk	socio- demographic	emergency medicine
cardiology	text	chatbots	dermatology	triage
research	strategy	wearable devices	digital assistant	lab tests
EHR	speech	education	screening	robotics
scheduling	language translation	genetic	nuclear medicine	radiation therapy

Credit: Joshua Reed, PhD

Examples of problems and challenges to solve with AI

improve diagnoses	improve treatments	scheduling and logistics	better decisions	improve efficiency
faster diagnoses	faster treatments	risk stratification	patient privacy	reduce bias
reduce errors	reduce waiting times	communication	monitoring	
information retrieval	self-service and self-care	minimize administrative work	expand access	
improve patient safety	optimize resource usage	improve research	patient preparation	

4

DISCUSS ENHANCED PATIENT CARE WITH AI

Cardiology

patterns in 12-lead ECGs

- low ejection fraction
- atrial fibrillation
- hypertrophic cardiomyopathy
- cardiac amyloidosis
- aortic valve stenosis



An artificial intelligence-enabled ECG algorithm for the identification of patients with atrial fibrillation during sinus rhythm: a retrospective analysis of outcome prediction



Zachi I Attia, Peter A Noseworthy*, Francisco Lopez-Jimenez, Samuel J Asirvatham, Abhishek J Deshmukh, Bernard J Gersh, Rickey E Carter, Xiaoxi Yao, Alejandro A Rabinstein, Brad J Erickson, Suraj Kapa, Paul A Friedman*

Summary

Background Atrial fibrillation is frequently asymptomatic and thus underdetected but is associated with stroke, heart failure, and death. Existing screening methods require prolonged monitoring and are limited by cost and low yield. We aimed to develop a rapid, inexpensive, point-of-care means of identifying patients with atrial fibrillation using machine learning.

Lancet 2019; 394: 861–67

Published **Online**

August 1, 2019

[http://dx.doi.org/10.1016/](http://dx.doi.org/10.1016/S0140-6736(19)31721-0)

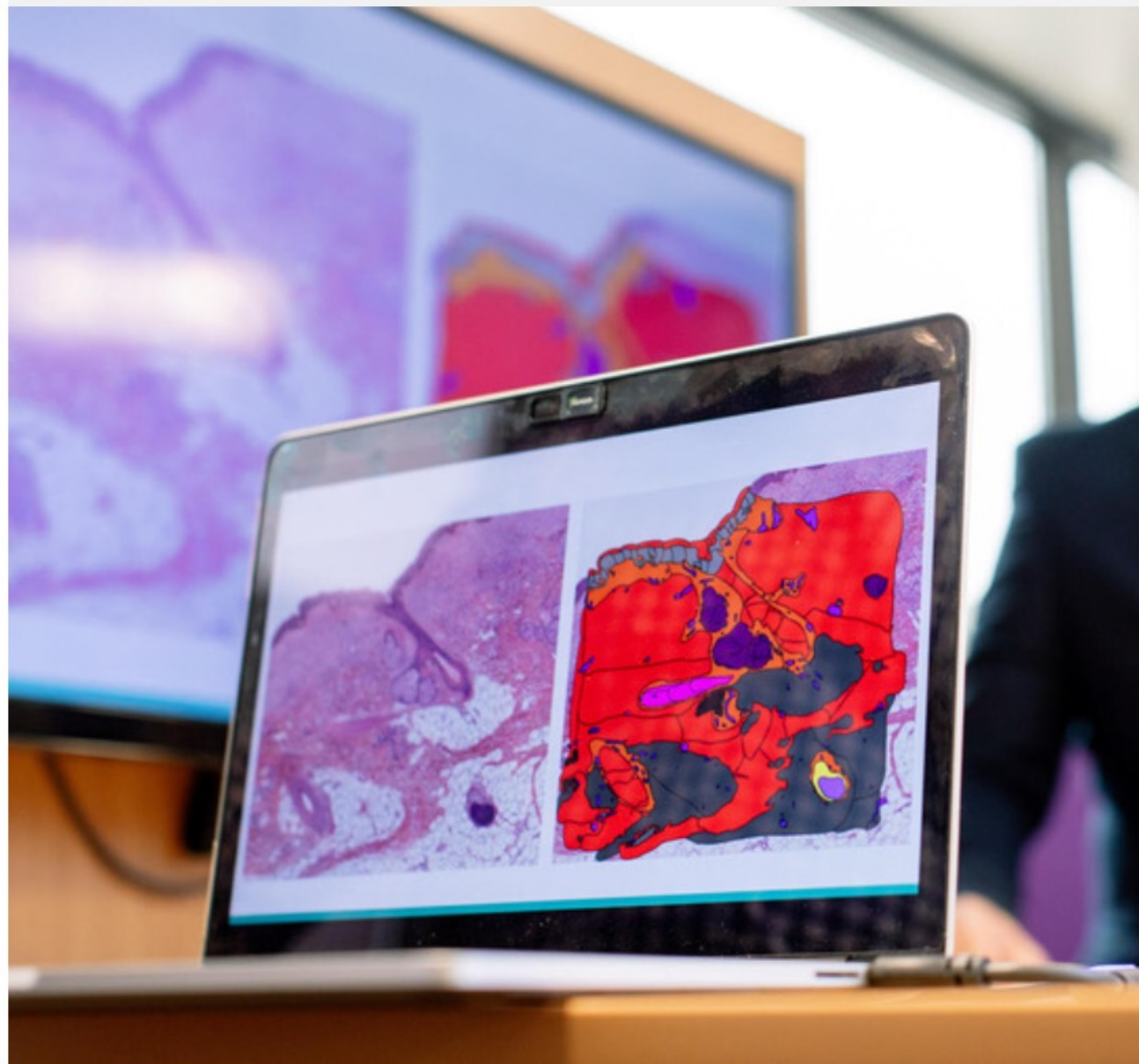
[S0140-6736\(19\)31721-0](http://dx.doi.org/10.1016/S0140-6736(19)31721-0)

[Attia, 2019, Lancet](#)

Computational Pathology

patterns in images

- highlight regions of interest in a slide
- detect presence of cancer
- tumor grading
- identify similar cases
- predict mutation status



Impact of Artificial Intelligence on Miss Rate of Colorectal Neoplasia

Michael B Wallace¹, Prateek Sharma², Pradeep Bhandari³, James East⁴, Giulio Antonelli⁵, Roberto Lorenzetti⁶, Micheal Vieth⁷, Ilaria Speranza⁸, Marco Spadaccini⁶, Madhav Desai³, Frank J Lukens⁹, Genci Babameto¹⁰, Daisy Batista¹⁰, Davinder Singh¹⁰, William Palmer⁹, Francisco Ramirez¹¹, Rebecca Palmer⁴, Tisha Lunsford¹¹, Kevin Ruff¹¹, Elizabeth Bird-Liebermann⁴, Victor Ciofoaia¹⁰, Sophie Arndtz³, David Cangemi⁹, Kirsty Puddick³, Gregory Derfus¹², Amitpal S Johal¹³, Mohammed Barawi¹⁴, Luigi Longo¹⁵, Luigi Moro¹⁵, Alessandro Repici¹⁶, Cesare Hassan¹⁶

	AI first (n = 116)	Standard colonoscopy first (n=114)	P value	OR [95% CI]
Subjects with ≥ 1 colorectal polyp in the 2 nd colonoscopy, n (%)	33 (28.45)	55 (48.25)	.002	0.43 [0.25–0.74]
Subjects with ≥ 1 adenoma or carcinoma in the 2 nd colonoscopy, n (%)	29 (25.00)	52 (45.61)	.001	0.40 [0.23–0.70]
Subjects with ≥ 1 adenoma or carcinoma in the 2 nd colonoscopy in subjects with a negative 1 st colonoscopy, n (%)	3 (6.82%)	13 (29.55)	.006	0.17 [0.05–0.67]

Polyp and Adenoma/Carcinoma Detection Rates in the Second Colonoscopy (Per Patient Analysis): FAS Population

Paper medical records

patterns in text

- identify text in scanned images
- find specific medical information
- reduce the amount of time to review
 - piloted across 7 practice areas
 - 10 to 20 percent time savings!



Clinical narratives

patterns in text

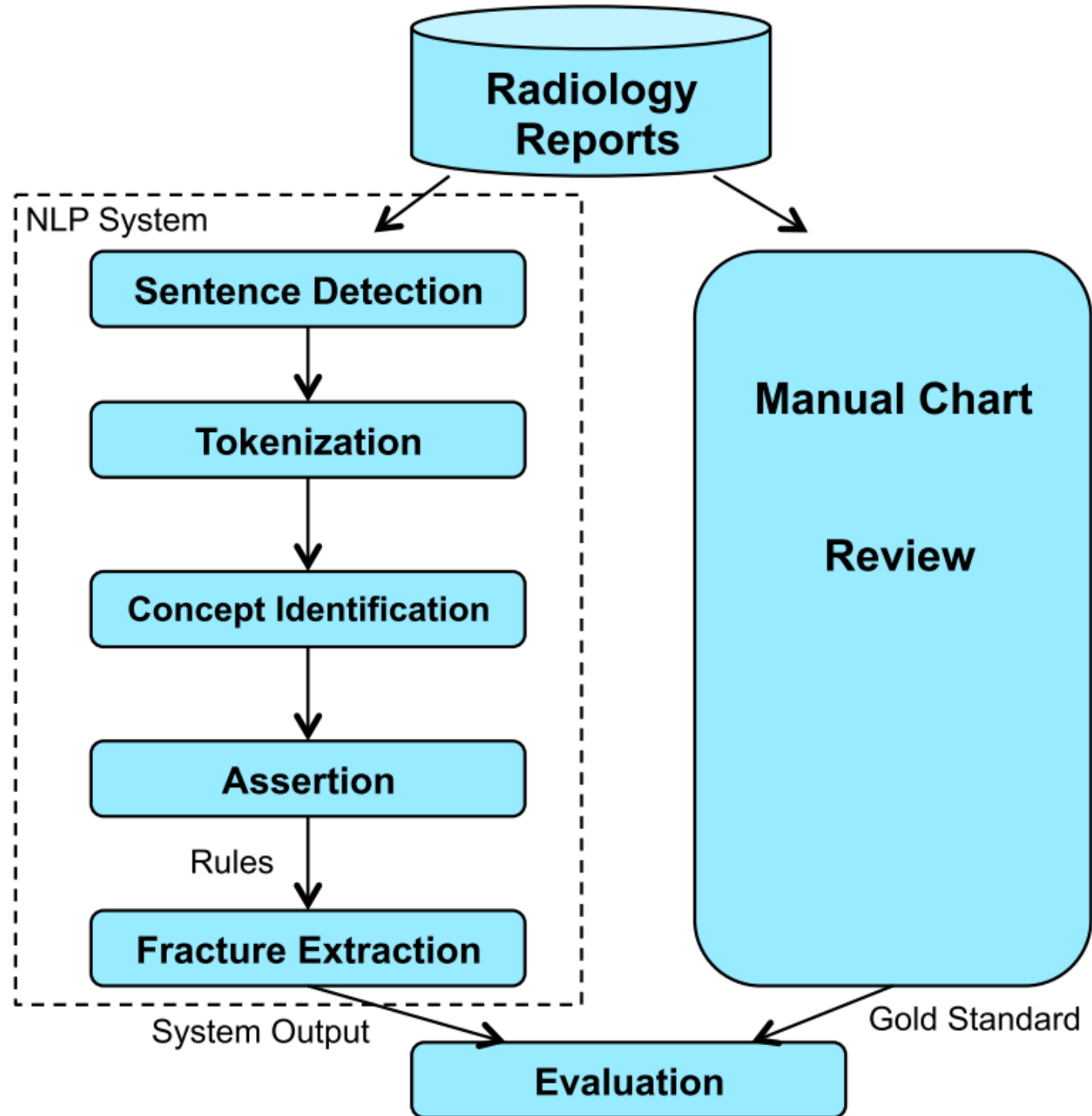
- radiology reports, clinical history, physical exam findings, operative reports
- detect keywords and concepts
- extract information
- appendicitis, pneumonia, potentially malignant lesions, skeletal site-specific fractures, asthma status, allergy status

natural language processing (NLP)

Exam Number: [REDACTED] Date: [REDACTED] Time: [REDACTED] Type: [REDACTED] Exam Code: [REDACTED] Ordering Provider: [REDACTED]	Exam Number: [REDACTED] Date: [REDACTED] Time: [REDACTED] Type: [REDACTED] Exam Code: [REDACTED] Ordering Provider: [REDACTED]
HISTORY [REDACTED] [REDACTED]	HISTORY [REDACTED] [REDACTED]
COMPARISON [REDACTED]	COMPARISON [REDACTED]
FINDINGS [REDACTED] [REDACTED] [REDACTED] [REDACTED]	FINDINGS [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
IMPRESSION [REDACTED] [REDACTED]	IMPRESSION [REDACTED] [REDACTED]
RECOMMENDATION [REDACTED]	RECOMMENDATION [REDACTED]

NATURAL LANGUAGE PROCESSING

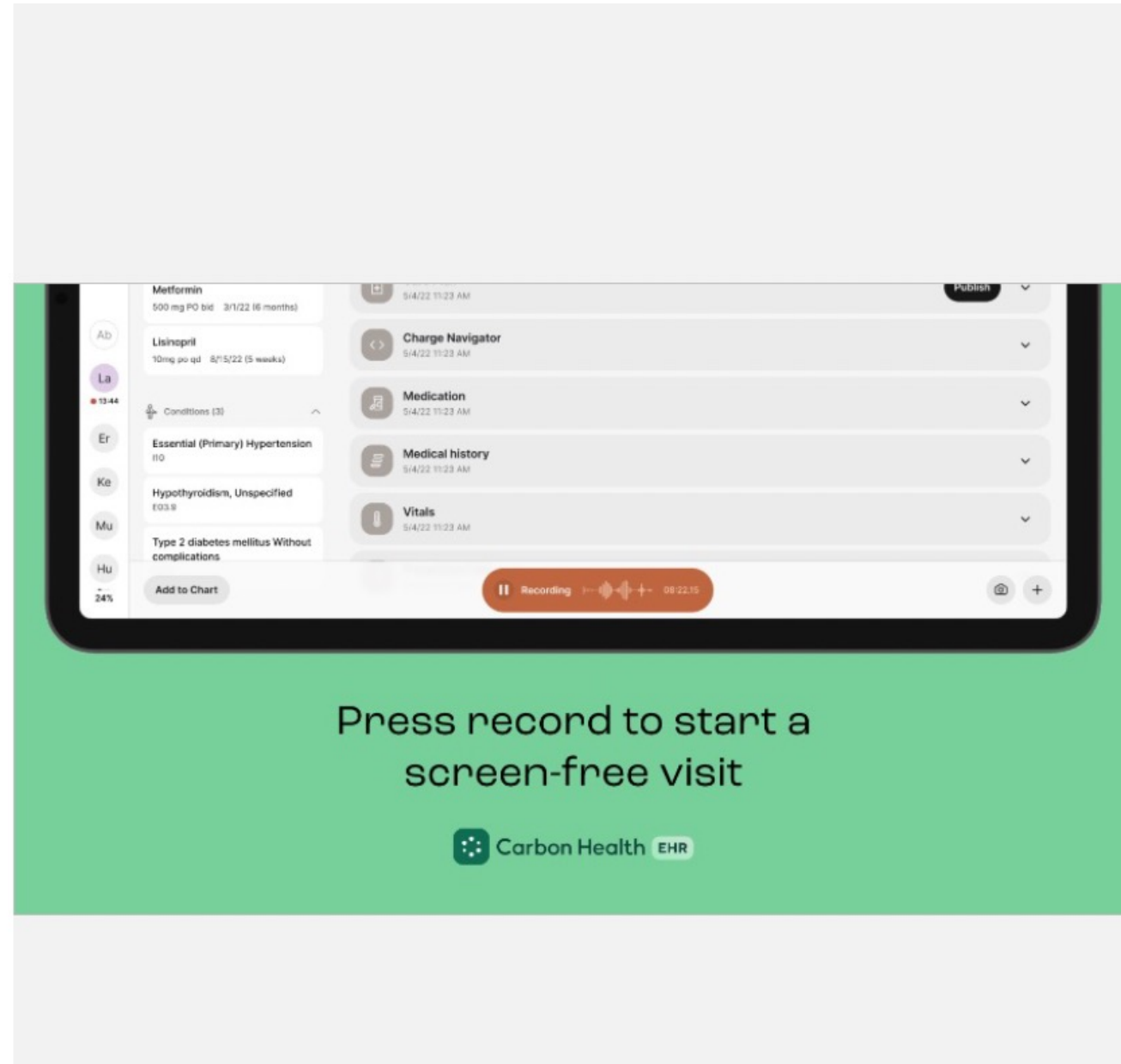
- An NLP algorithm was developed to identify twenty specific skeletal site fractures from radiology reports.
- Empirical experiments were conducted to validate the effectiveness of the NLP algorithm using radiology reports from a community-based cohort at Mayo Clinic.
- Microaveraged results of the NLP algorithm for the twenty fractures include sensitivity (0.930), specificity (1.0), PPV (positive predictive value, 1.0), NPV (negative predictive value, 0.941), and F1-score (0.961).



Charting

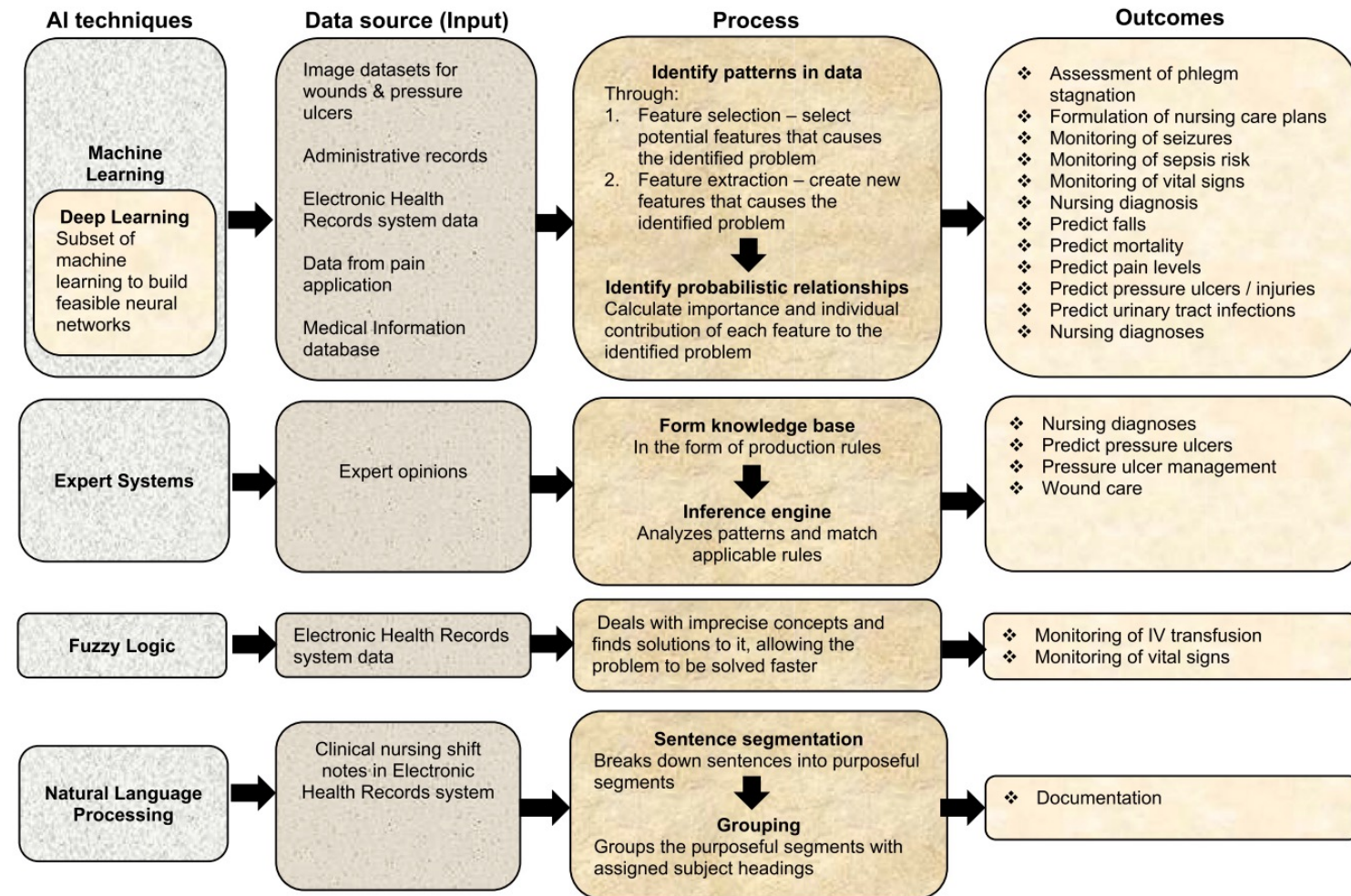
patterns in audio and EHR data

- conversation with patient is recorded
- recorded audio is transcribed
- transcript + EHR data → LLM
- LLM generates a visit summary
- *clinician reviews, edits, approves*



NURSING CARE

- AI-enhanced patient monitoring systems' efficiency, particularly for tracking vital signs, reduced nurses' workload, giving them more time for non-routine and less mentally taxing tasks.
- AI systems are trained to automatically monitor patient data and detect abnormalities, especially in critical care situations. Timely alerts can help in managing patients effectively.
- Limited real-world implementation



Diagrammatic summary of the mechanism by which artificial intelligence (AI) improves nursing care

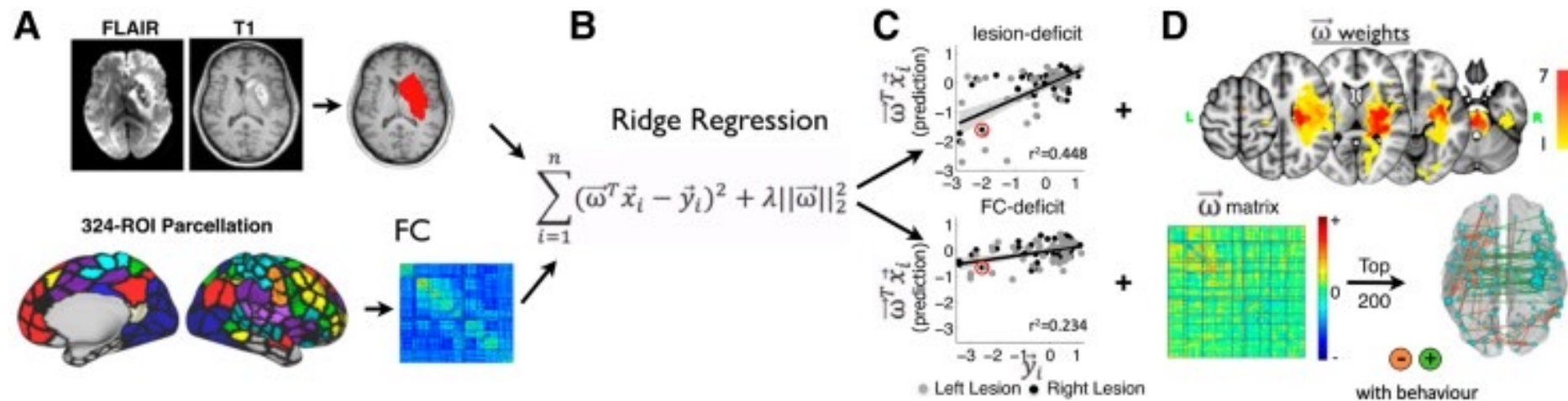
Precision medicine in stroke: towards personalized outcome predictions using artificial intelligence

Anna K Bonkhoff¹, Christian Grefkes^{2 3 4}

Affiliations + expand

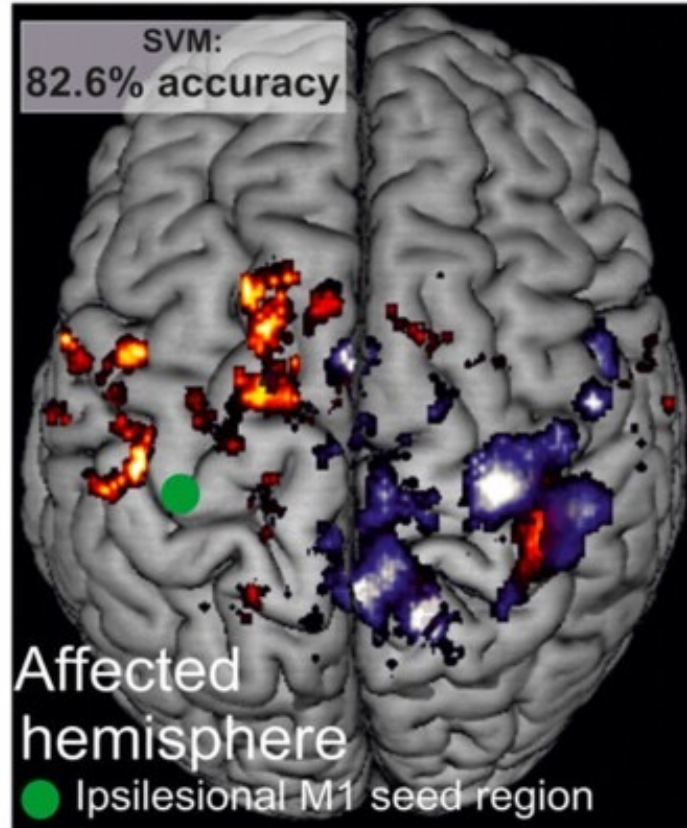
PMID: 34918041 PMCID: PMC9014757 DOI: 10.1093/brain/awab439

[Free PMC article](#)

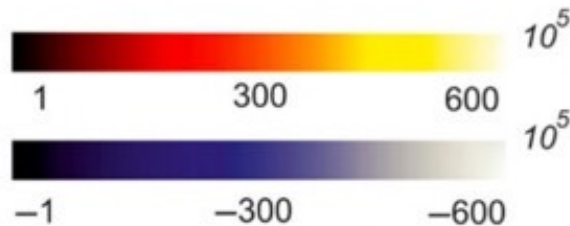
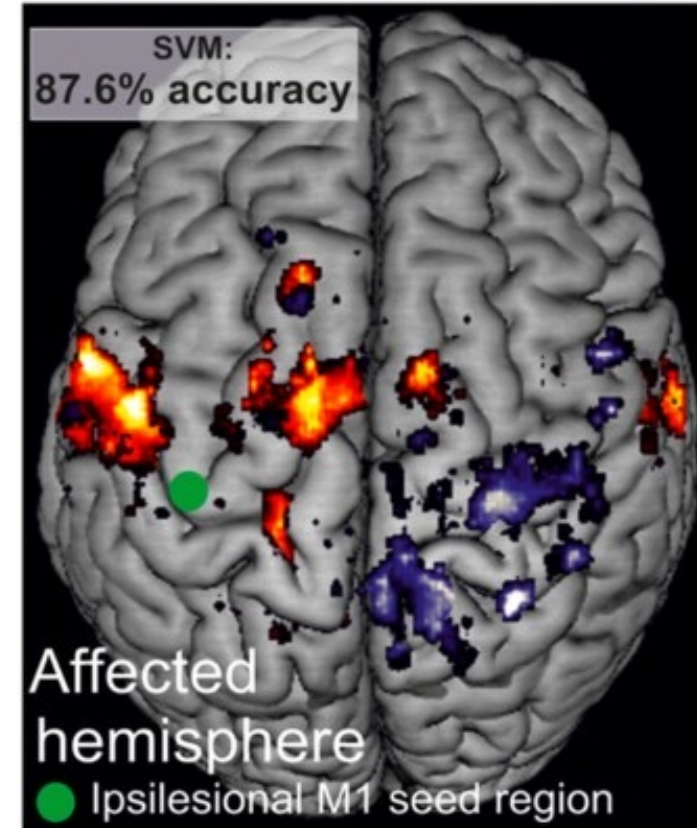


Overview of the analytical pipeline to predict behavioral impairments in 100 stroke patients based on structural and functional MRI.

A SVM:
Stroke patients with motor deficits
versus non-stroke controls



B SVM:
Stroke patients with versus
without hand motor deficits



DERMATOLOGY

- Teledermatology using AI offers dermatological expertise.
- Nondermatologist clinicians increase confidence and improve appropriateness of referrals to dermatologists.
- AI can reduce unnecessary referrals and enhance the detection of medically concerning ones, potentially expediting access to dermatologists.
- Dermatologists have a crucial role in ensuring that it is patient-centered, clinician-led, high-quality, equitable, and accessible.

Review > J Am Acad Dermatol. 2022 Dec;87(6):1336-1342. doi: 10.1016/j.jaad.2022.08.028.

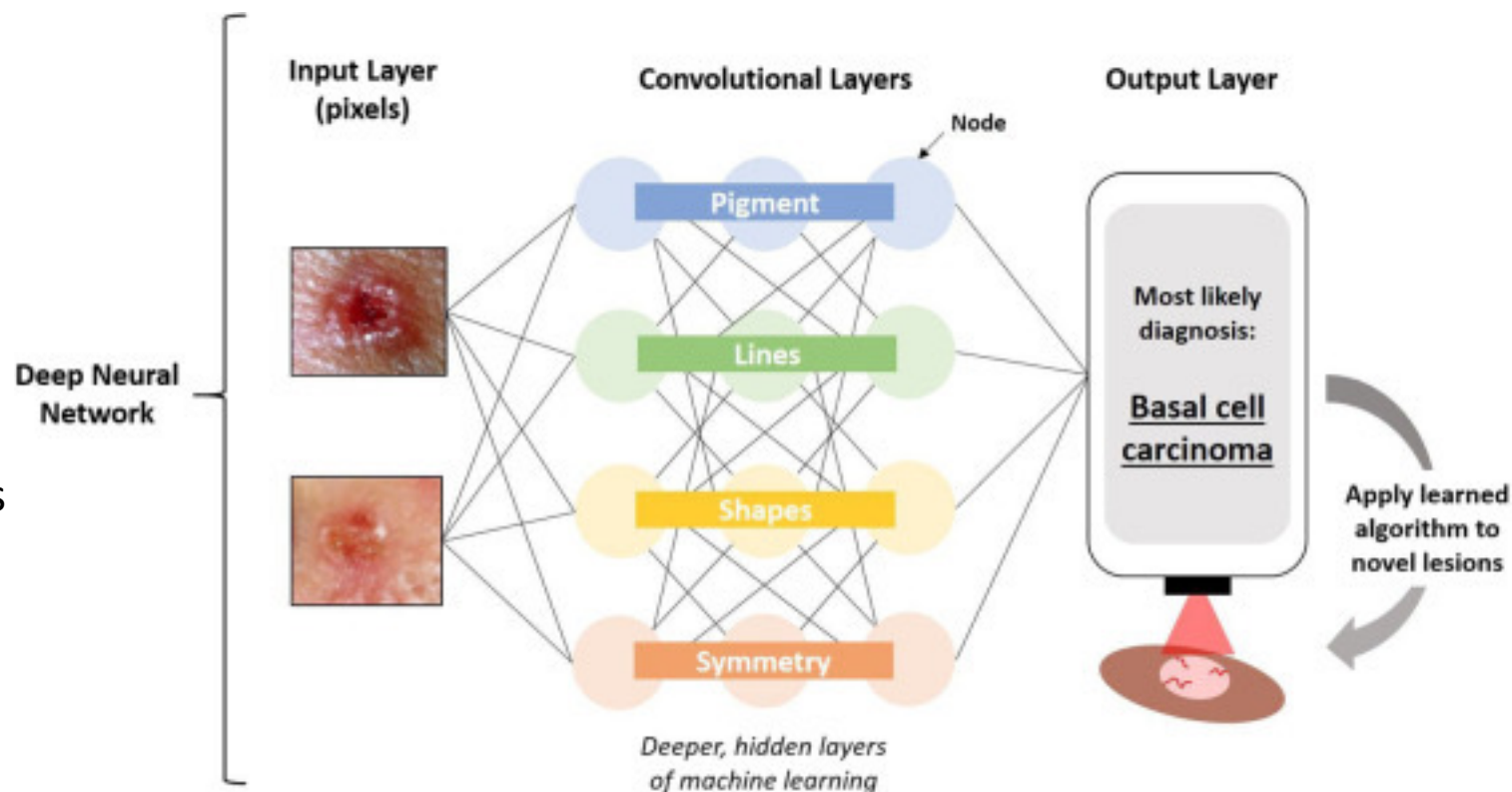
Epub 2022 Aug 23.

Artificial intelligence in the detection of skin cancer

Eric J Beltrami ¹, Alistair C Brown ², Paul J M Salmon ², David J Leffell ³, Justin M Ko ⁴, Jane M Grant-Kels ⁵

Affiliations + expand

PMID: 35998842 DOI: [10.1016/j.jaad.2022.08.028](https://doi.org/10.1016/j.jaad.2022.08.028)



Application of deep neural networks in the identification of skin neoplasms.

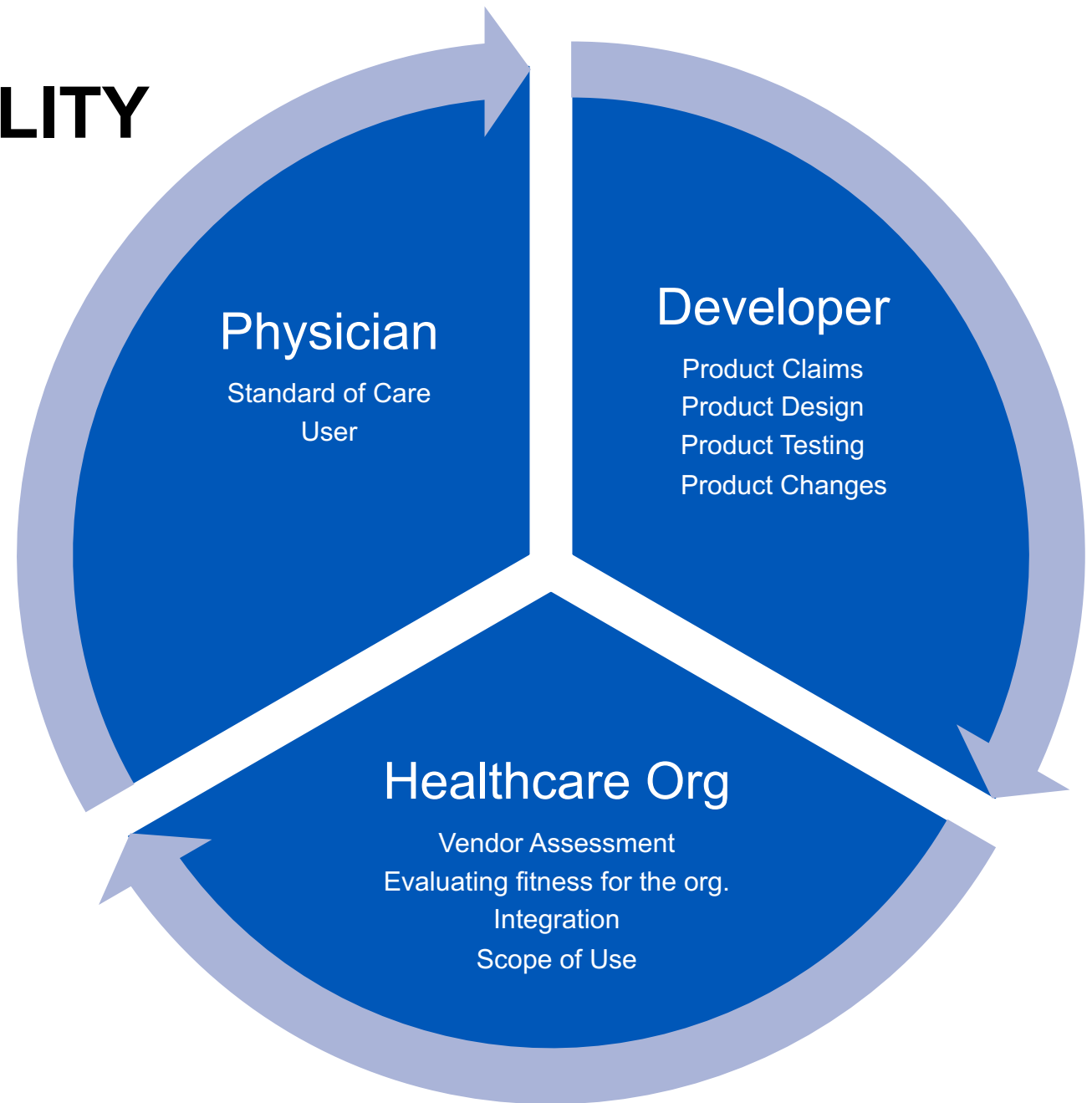


5

ANALYZE ETHICAL AND REGULATORY ASPECTS

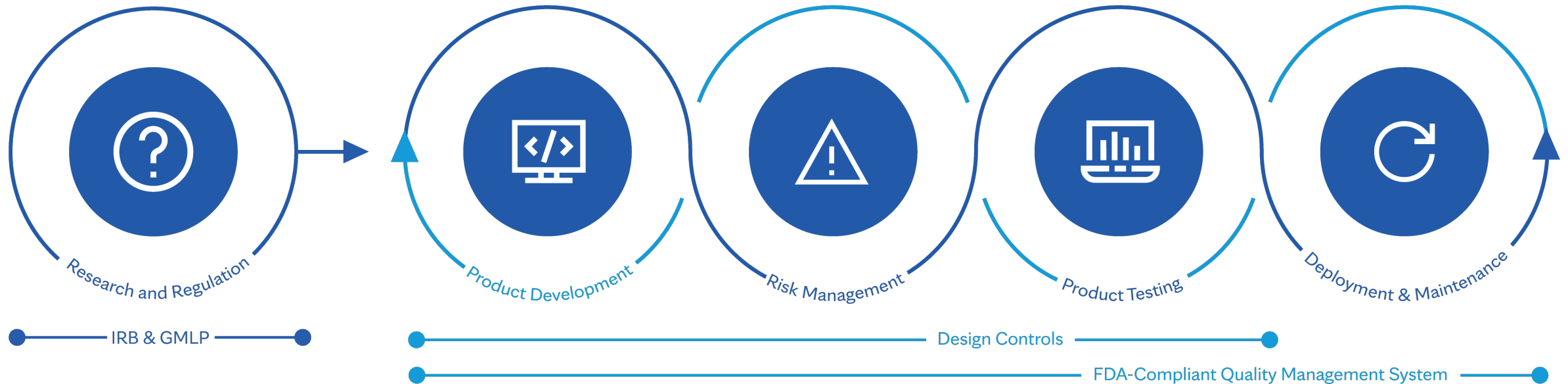
SHARED ACCOUNTABILITY

EXAMPLES

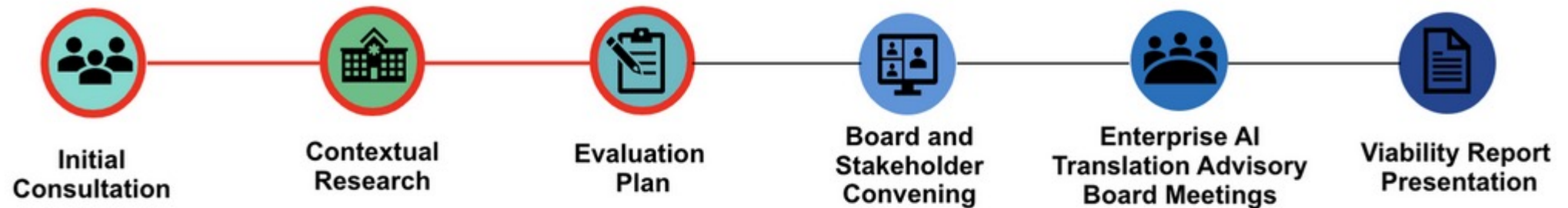


BUILDING A HEALTHCARE AI FRAMEWORK

HUB AND SPOKE MODEL

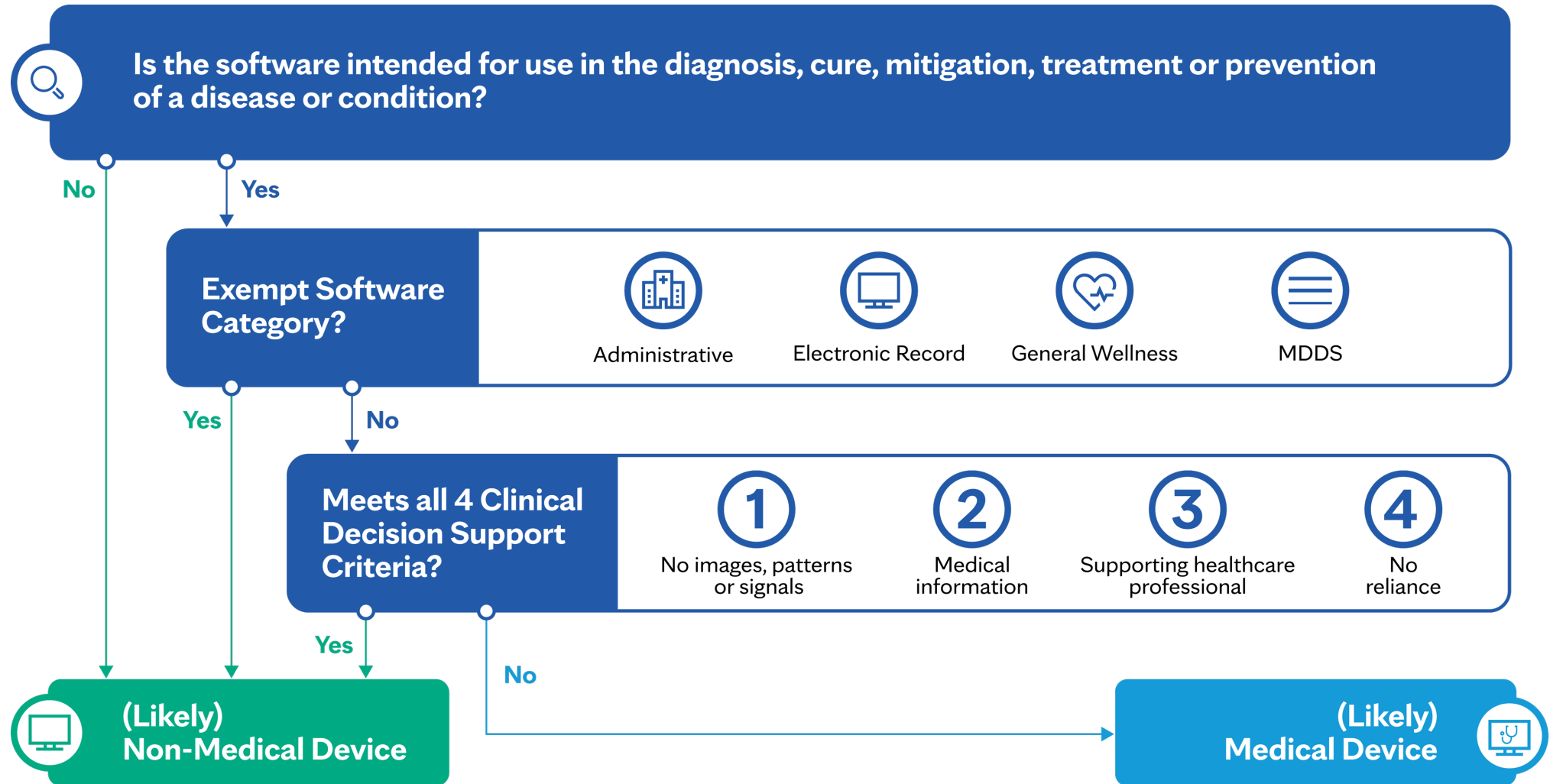


ENTERPRISE AI TRANSLATIONAL ADVISORY BOARD



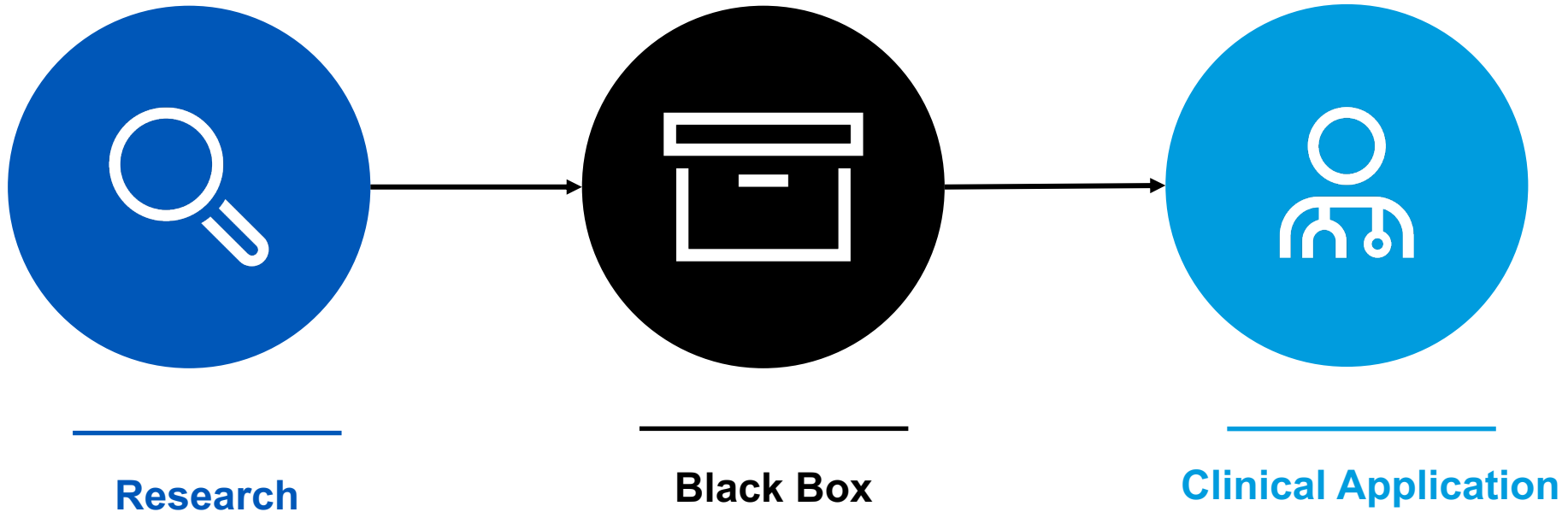
USA STANDARDS, IS YOUR PRODUCT A MEDICAL DEVICE?

ANALYZING EACH PRODUCT



TRANSLATING FROM RESEARCH TO PRACTICE

HEALTHCARE ORGANIZATIONS ENTERING NEW TERRITORY



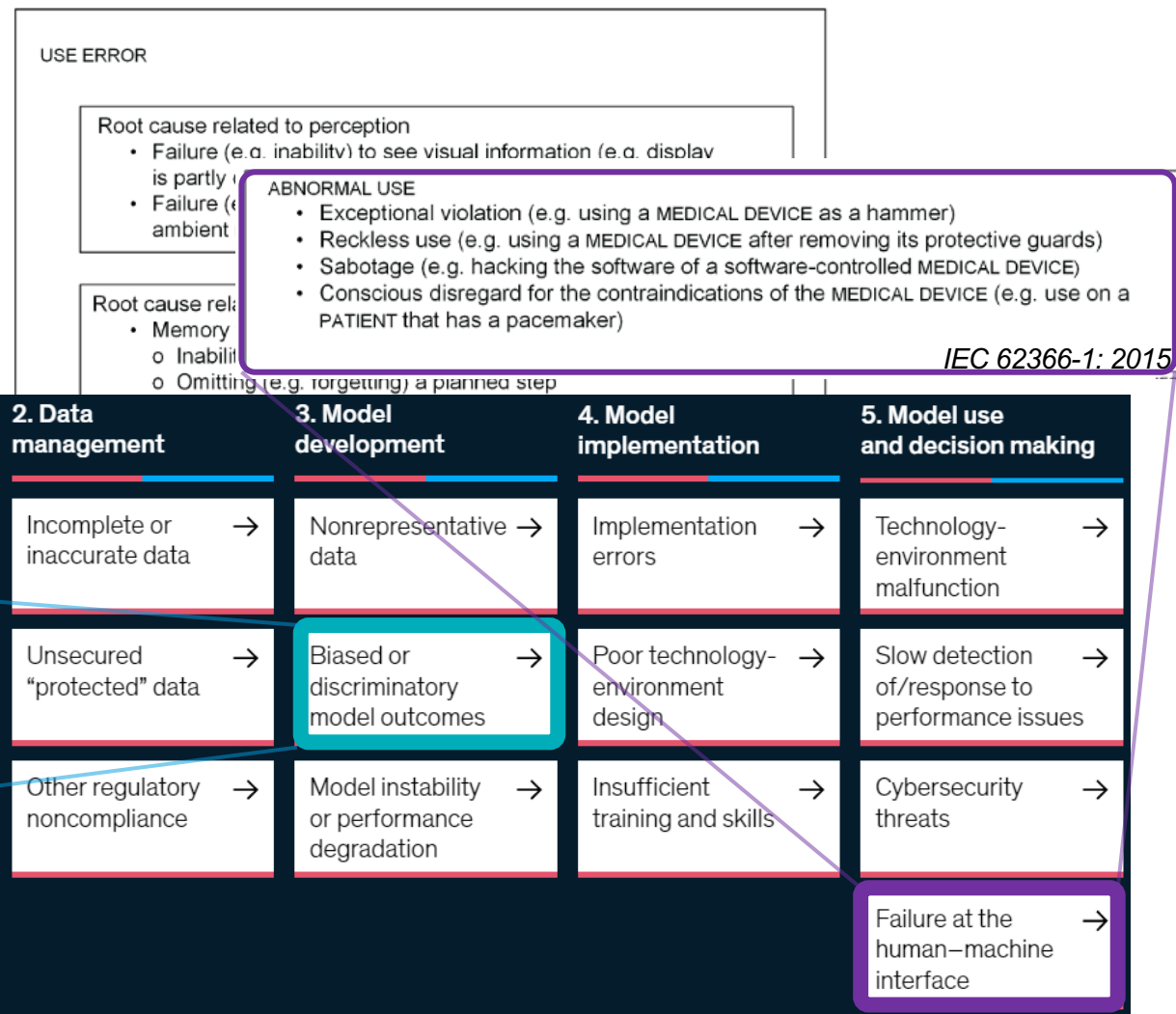
Credit: Mayo Clinic Center for Digital Health AI Enablement Team

AI/ML RISKS

Table B.1—Examples of ML-related hazards

Performance-related hazards	Data storage/security/privacy
Data quality	Privacy failures
Incorrect data	Inability to contact patient
Incorrect handling of outliers	Overtrust
Incomplete data	Overconfidence
Subjective data	Perceived risk
Inconsistent data	User workload
Atypical data	Self-confidence
Bias	Variation in social trust
Selection bias	Use policies
Confounding variables	
Non-normality	
Proxy variables	
Implicit bias	
Group attribution bias	
Experimental bias	

TIR34971:2023



<https://www.mckinsey.com/business-functions/quantumblack/our-insights/confronting-the-risks-of-artificial-intelligence>

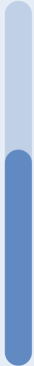
Evaluate and consider AI/ML risks for downstream impact

Credit: Mayo Clinic Center for Digital Health AI Enablement Team

©2022 Mayo Foundation for Medical Education and Research | slide-36

OUR DEPLOYMENT OPTIONS

BUILDING ON A FIRM FOUNDATION

				
	01 IRB Feasibility Testing & Research	02 Non-Medical Device	03 Translating Medical Device To Practice	04 FDA Submission & Commercialization
How is my product deployed?	Research Deployment	Non-Research Deployment	Non-Research Deployment	Non-Research Deployment
Who has primary oversight over my product?	Institutional Review Board (IRB)	Individual Department/Software Team	Internal Accountability Body (e.g., SaMD Review Board)	Food & Drug Administration (FDA)
Can I market or promote my product externally?	No	Yes	No	Yes
What risk-based controls must I put into place?	 Level of Effort IRB Controls	 Level of Effort IRB Controls Risk-Based Design Controls	 Level of Effort IRB Controls Risk-Based Design Controls Quality Management System	 Level of Effort IRB Controls Risk-Based Design Controls Quality Management System All FDA Controls



MAYO CLINIC SAMD REVIEW BOARD

SAFE, EFFECTIVE, AND ETHICAL
SOFTWARE FOR OUR PATIENTS



Regulatory Determinations

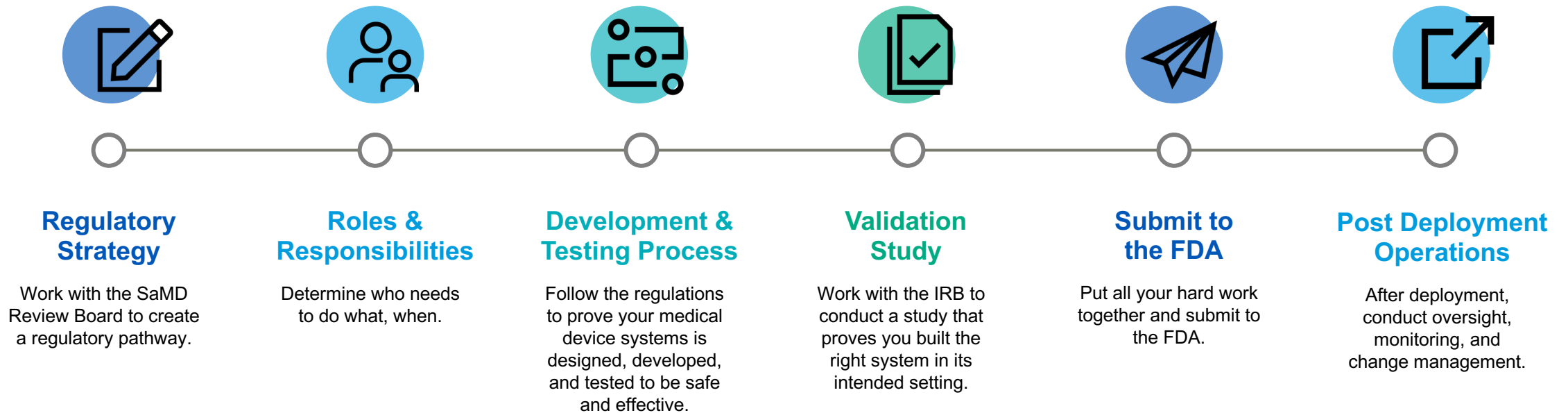


Risk Assessment

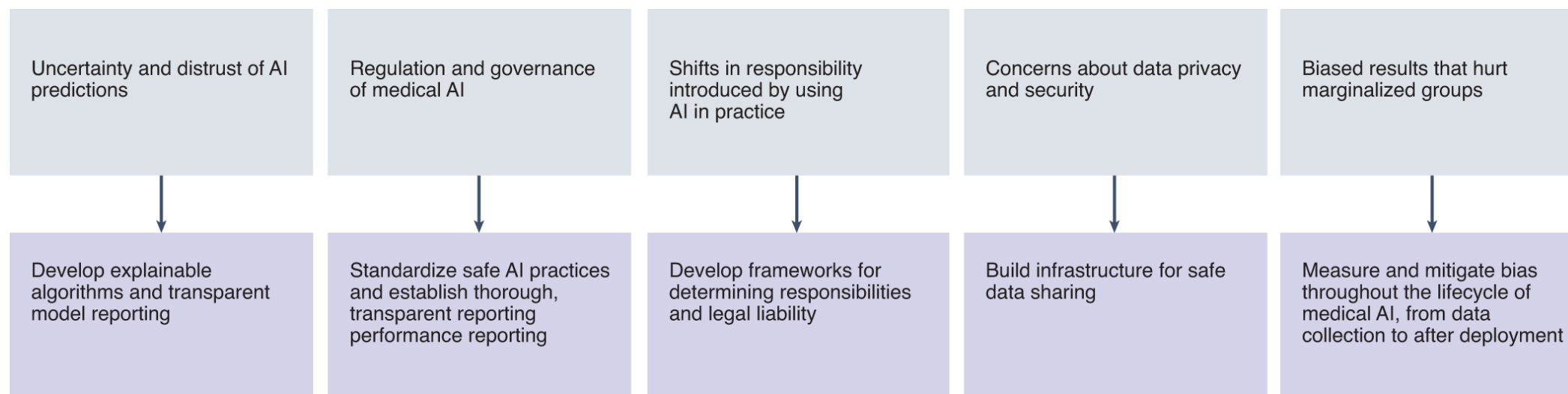


Recommended Risk Controls

SOFTWARE AS A MEDICAL DEVICE

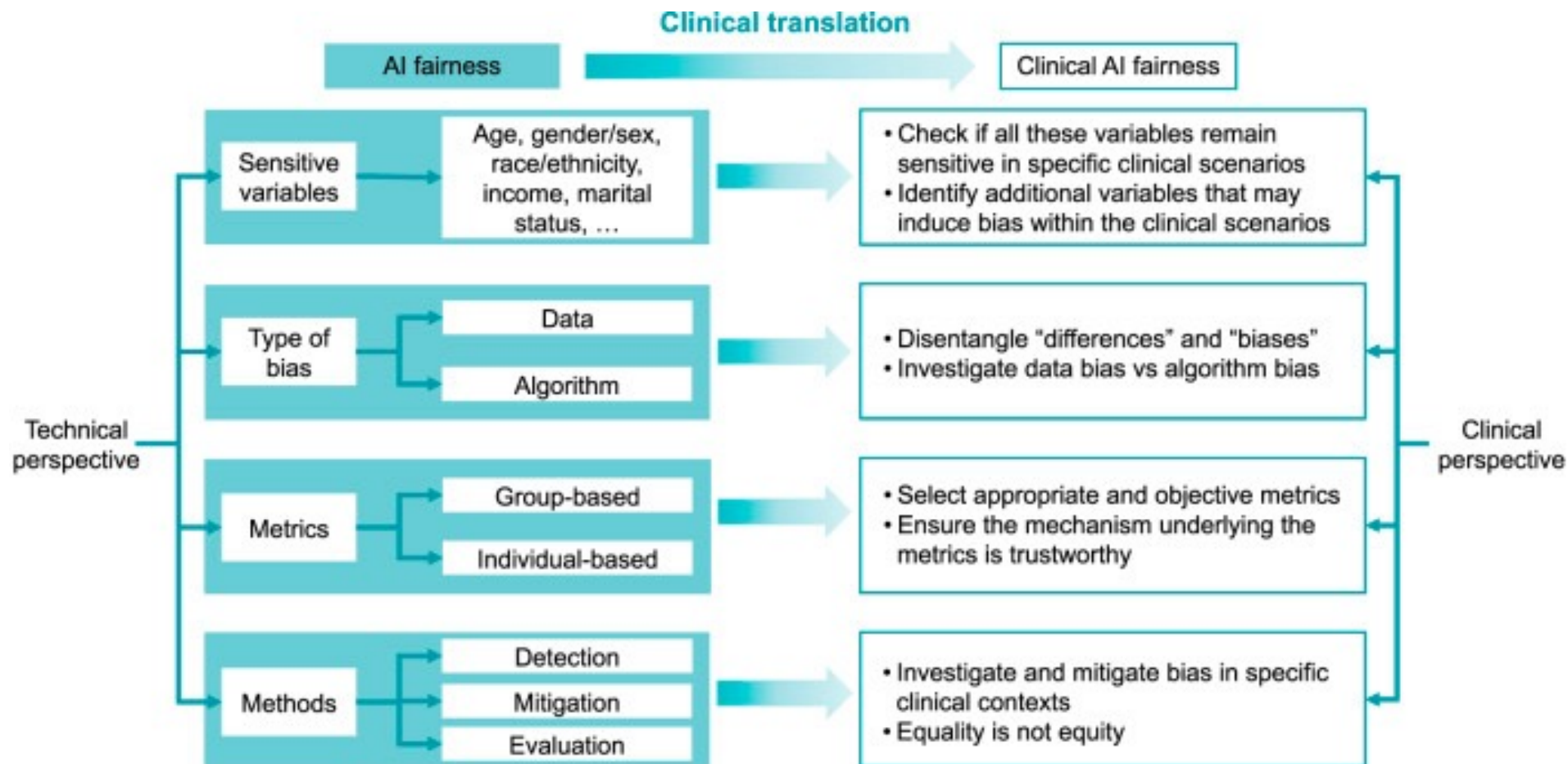


CHALLENGES



Ethical challenges for AI in medicine

AI FAIRNESS



Conceptual model towards clinical AI fairness.

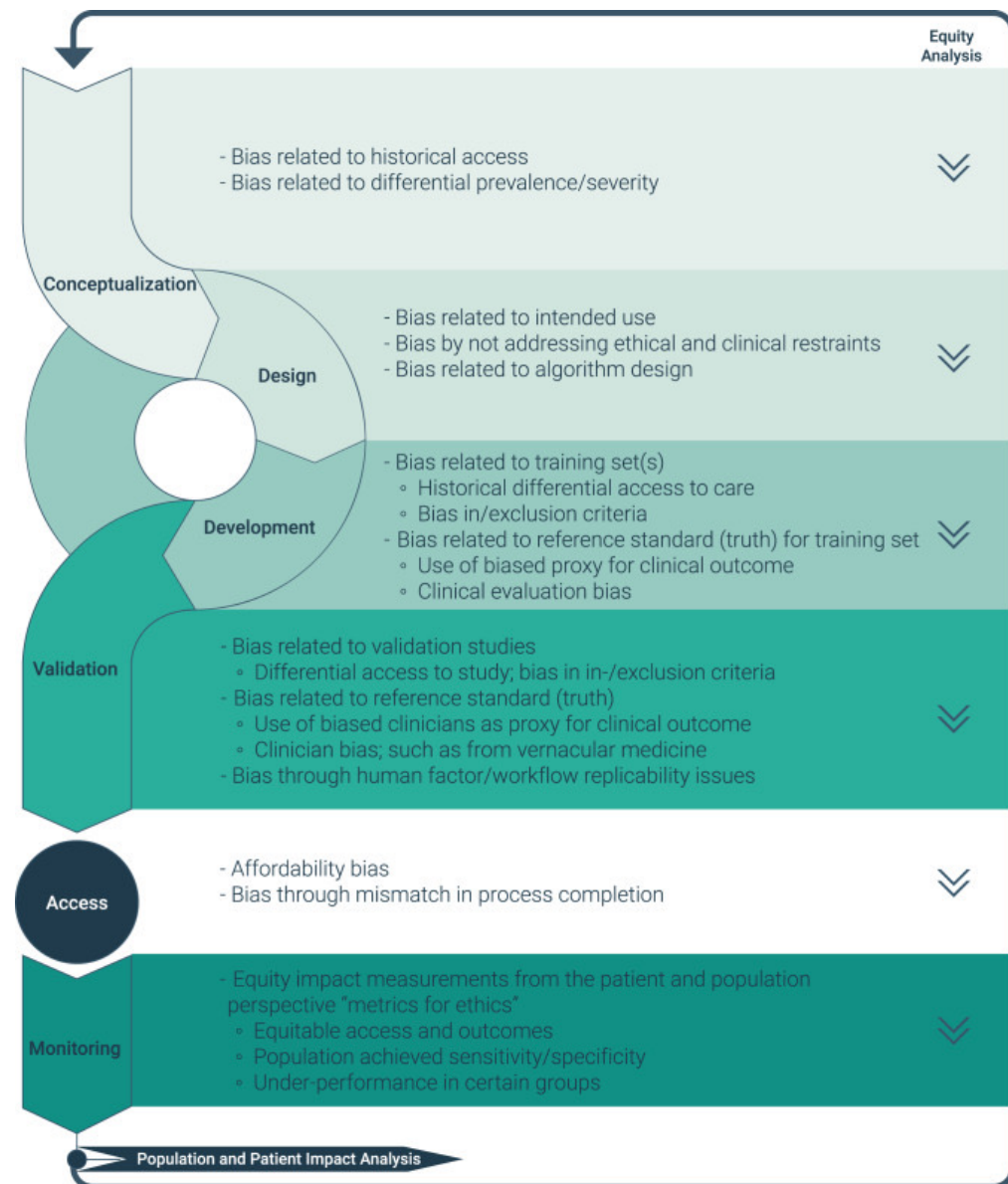
VULNERABLE POPULATIONS

AGE	- Include the patient's chronological age at the time of study enrollment - When applicable and available, include the patient's developmental age. If unavailable, state as such - Attempt to include developmental stages and relevant milestone metrics of CYP e.g., height and weight percentile upon enrollment, to capture the heterogeneity of participants. If developmental metrics are unavailable, state them as such - Include the age(s) of intended algorithm users e.g., pediatric only, pediatric and adult, or adult only
COMMUNICATION	- Communication of study purpose to CYP as key stakeholders with developmentally appropriate communication strategies - Communication with parent(s) or legal guardians as key stakeholders - Tailor communication to social circumstance, addressing family complexities, including any court involvement - Clear communication of technology-specific study purpose, risks, benefits, and alternatives with all key stakeholders - Consider the use of videos, written material, and decision aids to facilitate education and enhance communication - Involve stakeholders, including CYP and parents, in focus groups for design feedback where possible and relevant legal and institutional permissions are obtained - State efforts taken to involve potential users in feedback of research idea and invest in community-level digital literacy - Where possible, document and articulate model explainability
CONSENT AND ASSENT	- Record mode of consent, who provided consent, e.g., parent, legal guardian, and how it was obtained - Document any complex parental relations, dynamics, or court involvement that impact consent - Document children's social circumstances as relevant to safety, participation and evaluation - For children in state-custody, ensure consent is obtained by relevant legal guardians or custodians and documented accurately - Document relevant child protection laws pertinent to individual cases - Attain assent when developmentally appropriate and/or required by regulations - Ensure minors participate in the assent process in accordance with their developmental skills (e.g., appropriate modifications for children with clinically relevant developmental delay) - Record age when assent is provided - Ensure local laws for adolescent assent/consent are followed
EQUITY	- Ensure inclusion and exclusion criteria are clearly defined and specify disease, symptom, or condition of interest, with developmental stages considered as appropriate - State processes employed to reduce selection bias - Transparent demographic reporting, including race, documented sex, gender, and socioeconomic factors² - Provide details on how gender and documented sex have been incorporated into the study design² - Incorporate accessible research design to facilitate the inclusion of patients with disabilities (developmental and otherwise) - If skin tone could influence algorithmic outputs, ensure it is documented - Indicate the source of demographic information (e.g., self-reported) as well as details on non-reporting and missingness - Discuss the role of community engagement in the study
PROTECTION OF DATA	- State how data collection aligns with study objectives - State data-sharing plans when relevant - State if data is identifiable or de-identified - If data is de-identified, state compliance with the relevant legal frameworks, e.g., HIPAA, Common Rule, GDPR^{24,25} - State data protection plans, addressing unique data risks in AI/ML including protections against cybersecurity breaches. - Disclose whether data can or cannot be retrieved/removed in the future by parents and CYP - Ensure social context of child e.g., suspected or confirmed child abuse or complex social circumstance is accounted for prior to any data releases that may involve parental requests or involvement, if available
TECHNOLOGICAL CONSIDERATIONS - (TRANSPARENCY OF TECHNIQUES TRAINING, AND TESTING METHODOLOGY)	- Ensure algorithmic studies are tailored to the needs of the pediatric population and clearly documented in the study protocol - Ensure AI/ML techniques are only used when potentially beneficial to the pediatric population, and that such benefits are clearly detailed in the study protocol - Detail any potential harms that pediatric subjects may incur as a result of the study - Identify measures taken to minimize risk to pediatric subjects throughout study and post-implementation - State measures taken to monitor and document adverse events that may affect pediatric subjects - State outcome measures and plans to clinically evaluate performance of algorithms on pediatric subjects - When available, utilize validated pediatric clinical scales in the clinical algorithm evaluation - Articulate how AI/ML will be trained to recognize/account for developmental heterogeneity - Document AI/ML methods using validated guidelines, e.g CONSORT-AI and SPIRIT-AI^{10,11} - Define data input and output (e.g., images, text) as well as the source (e.g., public dataset), and output - Account for age-specific factors related to disability and developmental conditions (e.g., natural disease progression) as relevant in study design, testing, and evaluation - State if the study involves adult, pediatric, or mixed data in training and/or testing - If the study involves both adult and pediatric data, state the purpose for this combination - If the study involves both adult and pediatric data, state whether the same or separate algorithms were used to assess each group

ACCEPT-AI Framework: Key recommendations for pediatric data use in AI/ML research

AI BIAS

- Bias can originate at various stages of the TPLC and can be addressed using specific metrics and tailored approaches.
- Following an expanded TPLC approach, equity analysis and bias mitigation will help stakeholders understand how bias may affect healthcare decisions and outcomes.



Total Product LifeCycle (TPLC) equity expanded framework with examples for each phase.

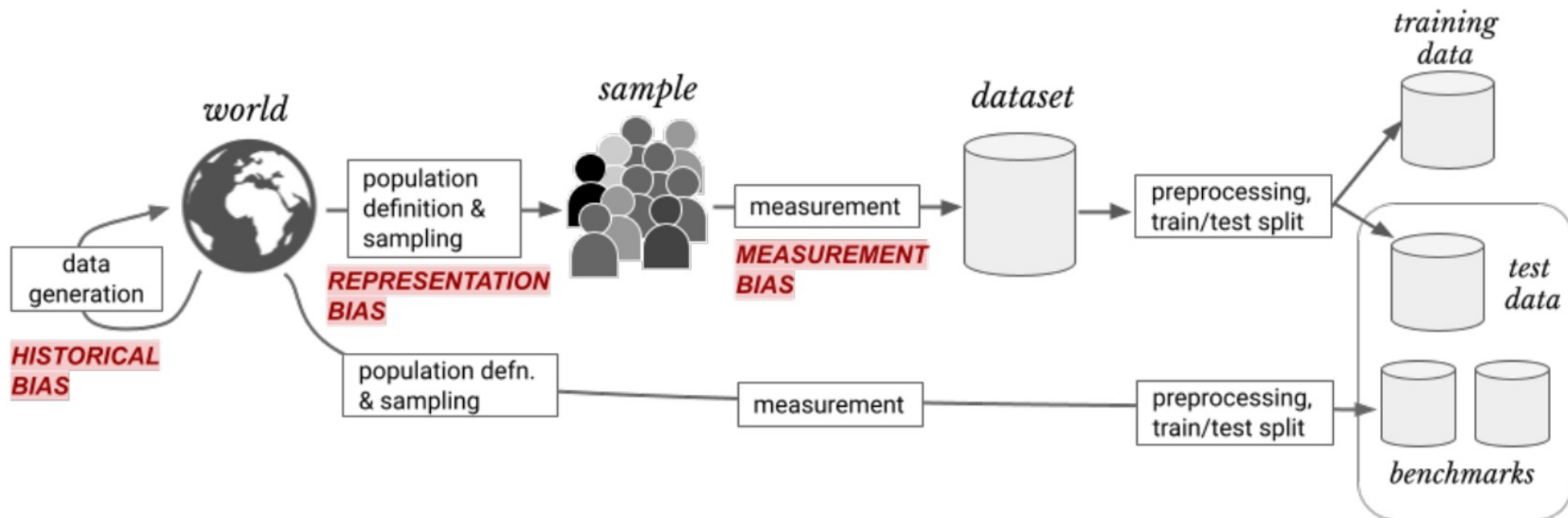
EXAMPLES OF AI BIAS SOURCES

- Biases in study inclusion criteria, e.g., using eGFR to select study participants
- Study data collection biases (including misclassification of race/ethnicity)
- Lack of representation / selection in the dataset
- Missing data
- Biases in imputed data
- Biases in learning and training data
- Collapsing race variables
- Lack of reporting for methodological approach
- Insufficient sample size
- Labeling bias
- Overfitting
- Lack of reporting for methodological approach
- Interpretation bias
- Correlation bias
- Training-validation data skew
- Lack of external validation
- Lack of performance assessments, such as calibration and discrimination
- Lack of reporting for methodological approach

DATA GENERATION

A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle

[Harini Suresh & John Gutttag. 2021.](#)



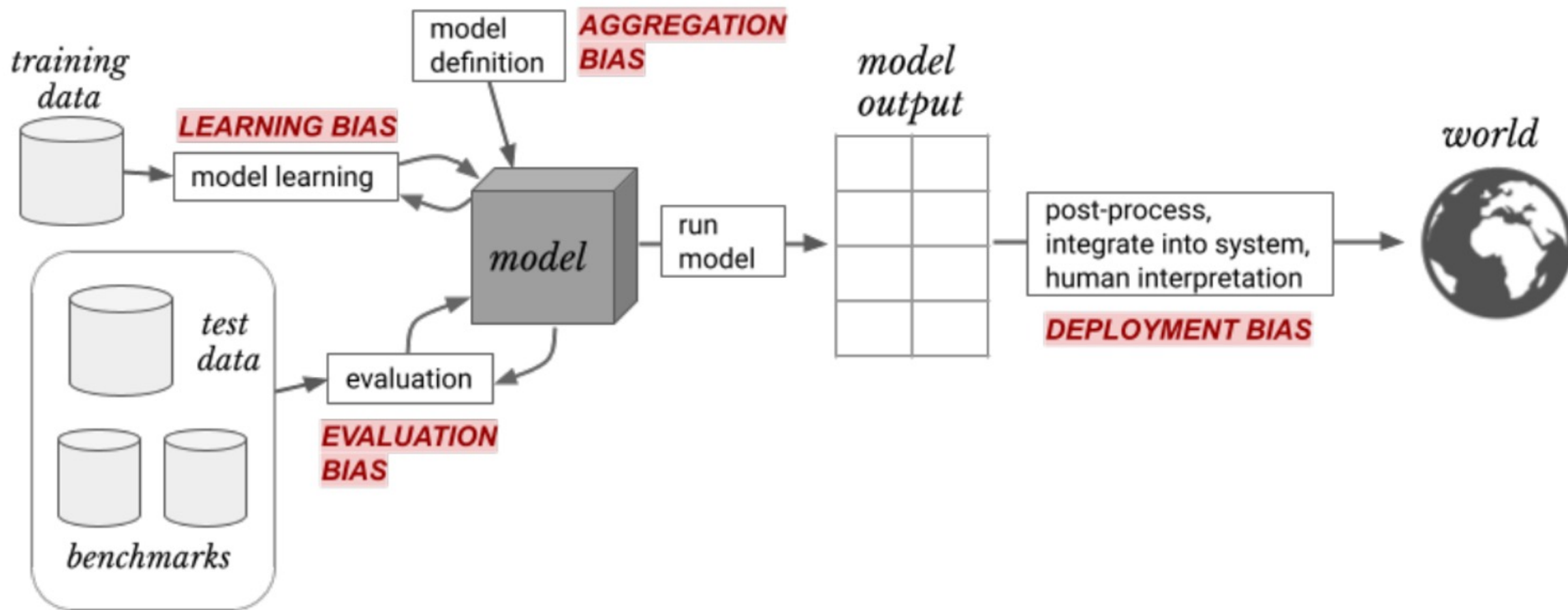
DATA GENERATION BIAS CONSIDERATIONS

Bias Type	Risk Description	Potential Mitigation	Resources
Historical/preexisting	Existing societal stereotypes reflected in datasets.	Debiasing Variational Autoencoder	Amini et al., 2019
Representation	Nonrepresentative data for population leading to systematic errors in ML model predictions.	Targeted data augmentation	Sontag & Johanssen, 2018
Measurement/cognitive	Chosen features and labels are imperfect proxies for the real variables of interest	Exchanging with domain experts about features	Baer 2019

MODEL BUILDING AND IMPLEMENTATION

A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle

[Harini Suresh & John Guttag. 2021.](#)



MODEL BUILDING AND IMPLEMENTATION BIAS CONSIDERATIONS

Bias Type	Risk Description	Potential Mitigation	Resources
Learning	Amplification of performance disparities	Reevaluate model design choices	Hooker, 2021
Aggregation	Incorrect assumption that mapping from inputs to labels is consistent across subsets of the data.	Coupled Learning Methods	Dwork et al., 2018
Evaluation	A non-representative testing population or inappropriate performance metrics are used to evaluate the ML model.	Representativeness of Benchmark Dataset Subgroup Validity	Giffen et al., 2021 Suresh & Guttag, 2019
Deployment	The ML model is used and interpreted in a different context than it was built for.	Holistic Evaluation Monitoring Plan & Human Supervision	Estiri et al., 2022 Giffen et al., 2022

AI EVALUATION: PROCESS STANDARDIZATION

Leveraging Clinical Research Pathway

Phase	Prepare: <i>0: Discovery and Intervention</i>	Develop: <i>1. Technical Performance and Safety</i>	Validate: <i>2. Efficacy and Side Effects</i>	Deploy: <i>3. Therapeutic Efficacy</i>	Maintain: <i>4. Safety and Effectiveness</i>
AI Evaluation Priorities	Data Quality Evaluation User-centered design work and user requirements specification Evaluation of user requirements for sustainability and scalability	Model Development SaMD Review; IRB and FDA regulatory determination Model ethical and explainability assessment UX design research and prototype design Deployment Feasibility Analysis PCED submits the endorsed EHR Enhancement Request AI Ethics & Bias Evaluation	Iterative interface design with stakeholder focus groups Silent Evaluation/Pre-pilot model performance and validation Model implementation evaluation and validation Business plan validation and deployment planning	Prospective Study/Pragmatic Trial Clinical Validation Operational Education	Post-deployment surveillance and maintenance risk assessment Quality monitoring and audit Model maintenance plan Corrective and preventative action model documentation Fidelity/surveillance of clinical utility
AI Evaluation and Stewardship Checkpoint	Patient Impact Clinical Practice and Business Impact Societal Impact Purpose and Indications Model Planning and Architecture Historical Bias Evaluation Representation Bias Evaluation	Risk Assessment Model Accuracy Evaluation Usability Formative Measurement Bias Evaluation Learning Bias Evaluation	Silent Evaluation Model Statistical Validation Deployment and Validation Planning Aggregation Bias Evaluation	Clinical Validation User Education and Training Plan Monitoring and Reporting Plan Corrective and Preventative Action Plan Deployment Bias Evaluation	Post-Deployment Maintenance Quality Monitoring and Audit Maintenance De-implementation Continual Bias Evaluation Plan

Artificial Intelligence Translation in Health Care: Framework for Evaluation and Documentation.

Overgaard, et al. [V2 in progress] V1 publication: <https://pubmed.ncbi.nlm.nih.gov/35854754/>



The Role of Artificial Intelligence Model Documentation in Translational Science: Scoping Review

Tracey A Brereton ¹ ; Momin M Malik ¹ ; Mark Lifson ¹ ; Jason D Greenwood ² ;
Kevin J Peterson ¹ ; Shauna M Overgaard ¹ 

Article

Authors

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Metrics

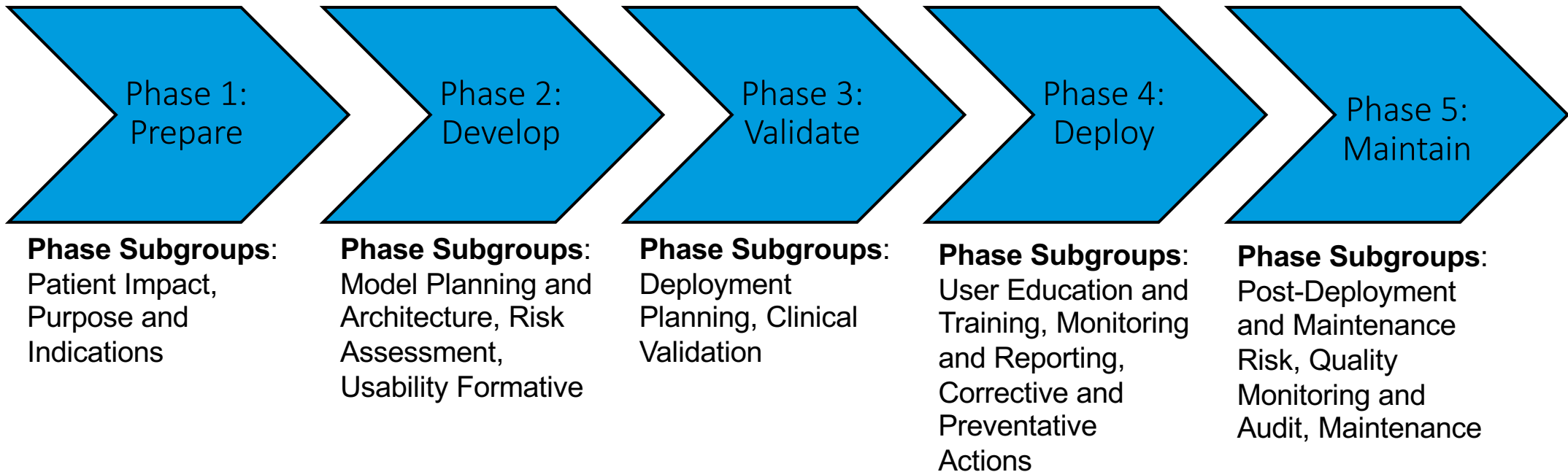
- [Abstract](#)
- Introduction
- Methods
- Results
- Discussion
- References
- Abbreviations
- Copyright

Abstract

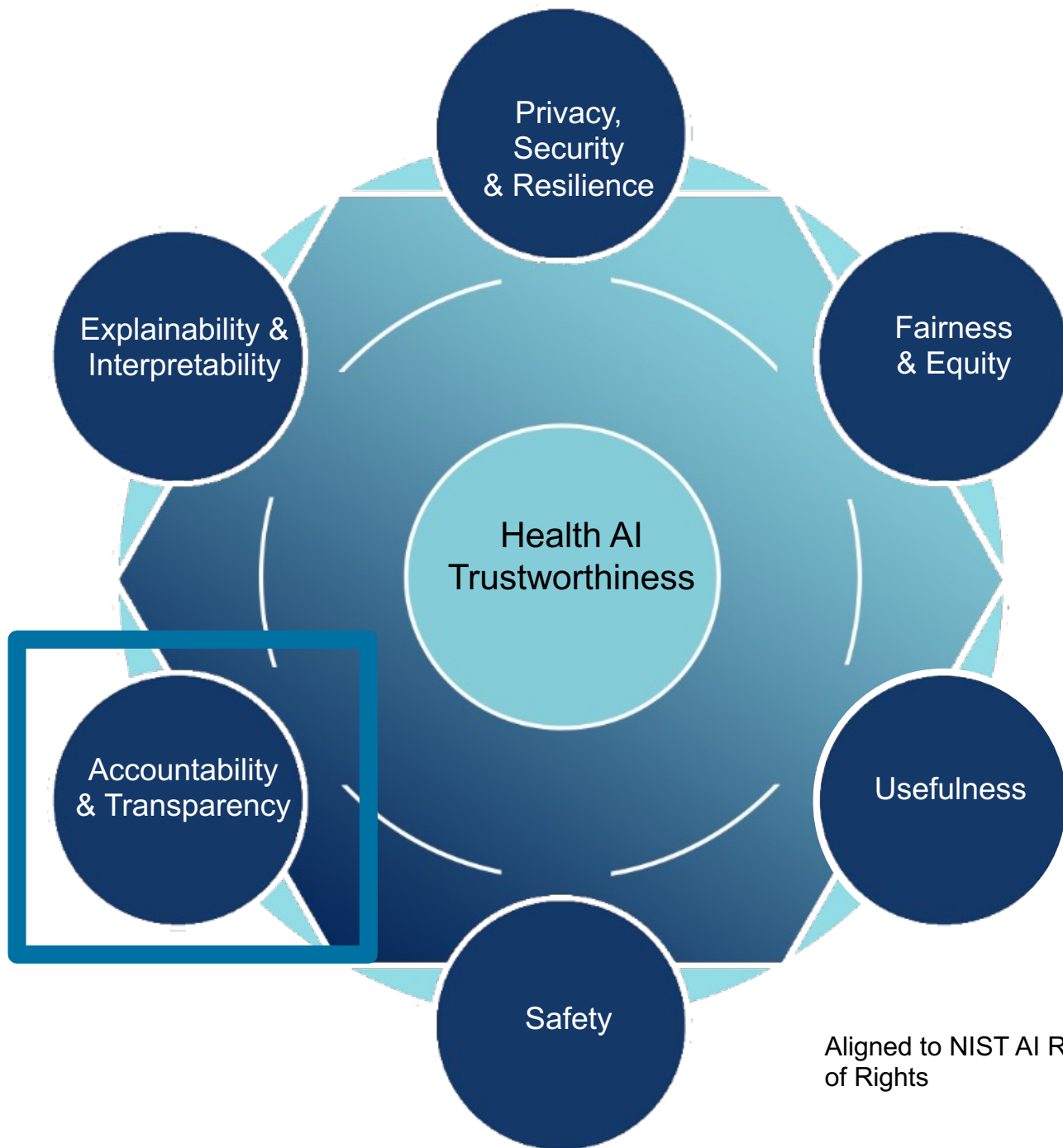
Background:

Despite the touted potential of artificial intelligence (AI) and machine learning (ML) to revolutionize health care, clinical decision support tools, herein referred to as medical modeling software (MMS), have yet to realize the anticipated benefits. One proposed obstacle is the acknowledged gaps in AI translation. These gaps stem partly from the fragmentation of processes and resources to support MMS transparent documentation. Consequently, the absence of transparent reporting hinders the provision of evidence to support the implementation of MMS in clinical practice, thereby serving as a substantial barrier to the successful translation of software from research settings to clinical practice.

MODEL DOCUMENTATION FRAMEWORK



Reporting Guidelines Used: CONSORT-AI, Model Card, TRIPOD | Updates include IEC 62304, ISO 14971



CORE CHAI/NIST ALIGNED PRINCIPLES

Aligned to NIST AI Risk Management Framework and the White House Blueprint for an AI Bill of Rights



HOW CAN WE ENSURE WE ARE ADVOCATING FOR THE SAFETY AND EFFICACY OF AI ON BEHALF OF OUR PATIENTS?



What policies, processes, procedures, and practices across the organization related to the mapping, measuring, and managing of AI risks are in place, transparent, and implemented effectively?



What accountability structures are in place so that the appropriate teams and individuals are empowered, responsible, and trained for mapping, measuring, and managing AI risks?



What are we doing to ensure that workforce diversity, equity, inclusion, and accessibility processes are prioritized in the mapping, measuring, and managing of AI risks throughout the lifecycle?



How are we building organizational teams that are committed to a culture that considers and communicates risk?

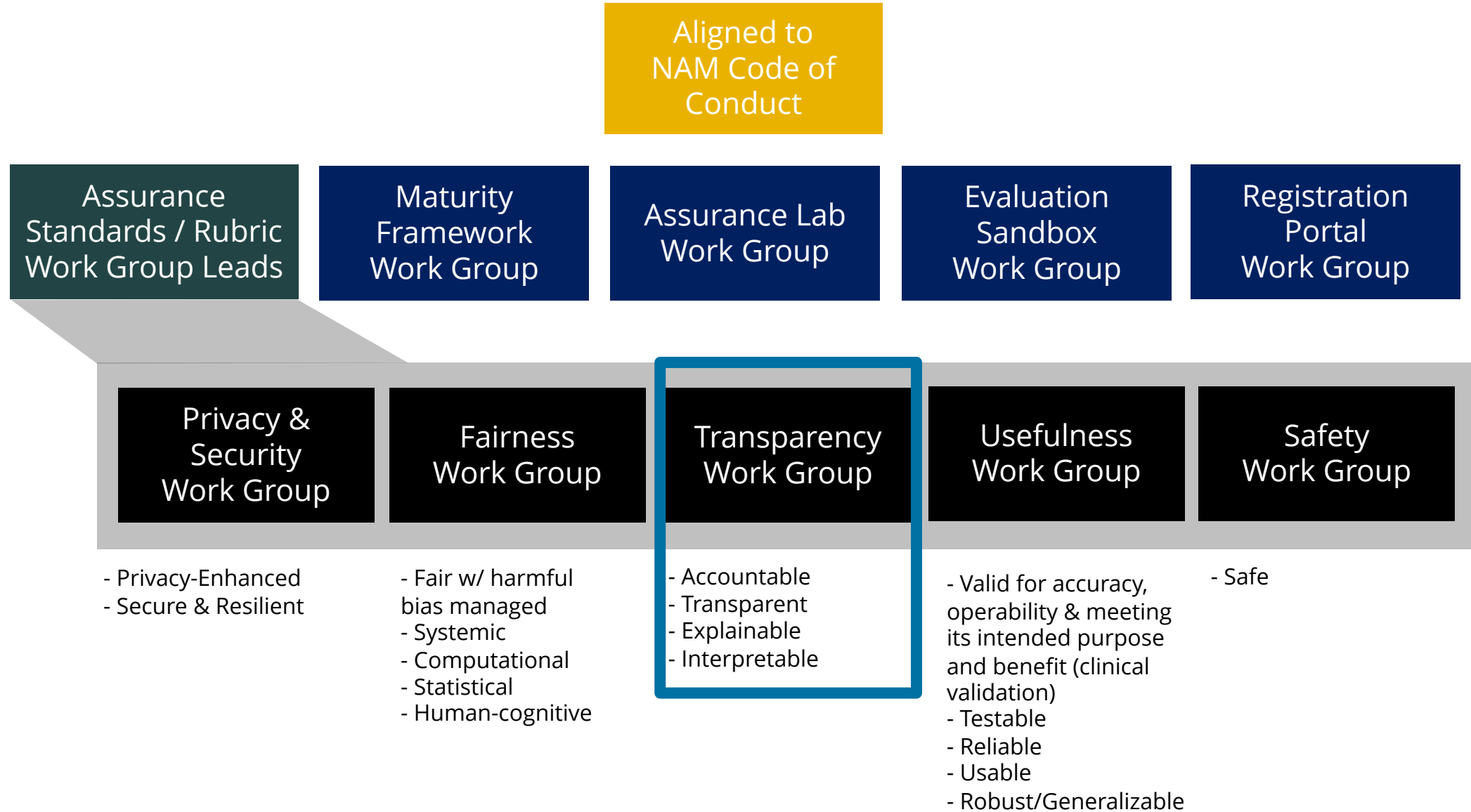


What processes are in place for robust stakeholder engagement?



Are there clear policies and procedures in place to address AI risks?

WORK GROUPS



CHAI
Community

The Global Digital Health Partnership

- Take a global approach to evidence building
- Sharing Insights and considering risk-based frameworks as a priority in digital health evaluation
- Evolve the workstream; move towards 'Evidence Translation and Implementation'



<https://gdhp.health/>



Comment

WHO and ITU establish benchmarking process for artificial intelligence in health

Thomas Wiegand ^a ✉, Ramesh Krishnamurthy ^b, Monique Kuglitsch ^a, Naomi Lee ^c, Sameer Pujari ^b, Marcel Salathé ^d, Markus Wenzel ^a, Shan Xu ^e

Focus Group on "Artificial Intelligence for Health"

YOU ARE HERE ITU > HOME > ITU-T > FOCUS GROUPS > ARTIFICIAL INTELLIGENCE FOR HEALTH

Focus Group on metaverse

Focus Group on Testbeds Federations for IMT-2020 and beyond

Focus Group on Artificial Intelligence (AI) and Internet of Things (IoT) for Digital Agriculture

Focus Group on AI for Natural Disaster Management

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FG-AI4H

The Focus Group on Artificial Intelligence for Health (FG-AI4H) is a partnership of ITU and the World Health Organization (WHO) to establish a standardized assessment framework for the evaluation of AI-based methods for health, diagnosis, triage or treatment decisions. Participation in the FG-AI4H is free and open to all. The group was established by ITU-T Study Group 16 at its meeting in Ljubljana, Slovenia, 9-20

Whitepaper for the ITU/WHO Focus Group on Artificial Intelligence for Health

Thomas Wiegand (Fraunhofer HHI, Germany), Naomi Lee (The Lancet, UK), Sameer Pujari (WHO), Manjula Singh (ICMR, India), Shan Xu (CAICT, China), Monique Kuglitsch (Fraunhofer HHI, Germany), Marc Lecoultre (MLLab.AI, Switzerland), Ana Riviere-Cinnamond (PAHO/WHO), Eva Weicken (Fraunhofer HHI, Germany), Markus Wenzel (Fraunhofer HHI, Germany), Alixandro Werneck Leite (University of Brasilia, Brazil), Simão Campos (ITU), and Bastiaan Quast (ITU)

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Abstract

Artificial intelligence (AI)—the phenomenon of machines being able to solve problems that traditionally require human intelligence—has seen an enormous rise in interest due to considerable advances in effectiveness and use. The health sector, one of the most important sectors for societies and economies worldwide, is particularly interesting for AI applications, given the ongoing digitalization of health data and the promise for an improved quality of health and healthcare. However, due to the complexity of AI models, it is difficult to distinguish good from bad AI-based solutions and to understand their strengths and weaknesses. This is crucial for clarifying

6

ANTICIPATE FUTURE AI TRENDS IN HEALTHCARE

FUTURE OPPORTUNITIES FOR AI IN HEALTHCARE



ADAPTIVE AI

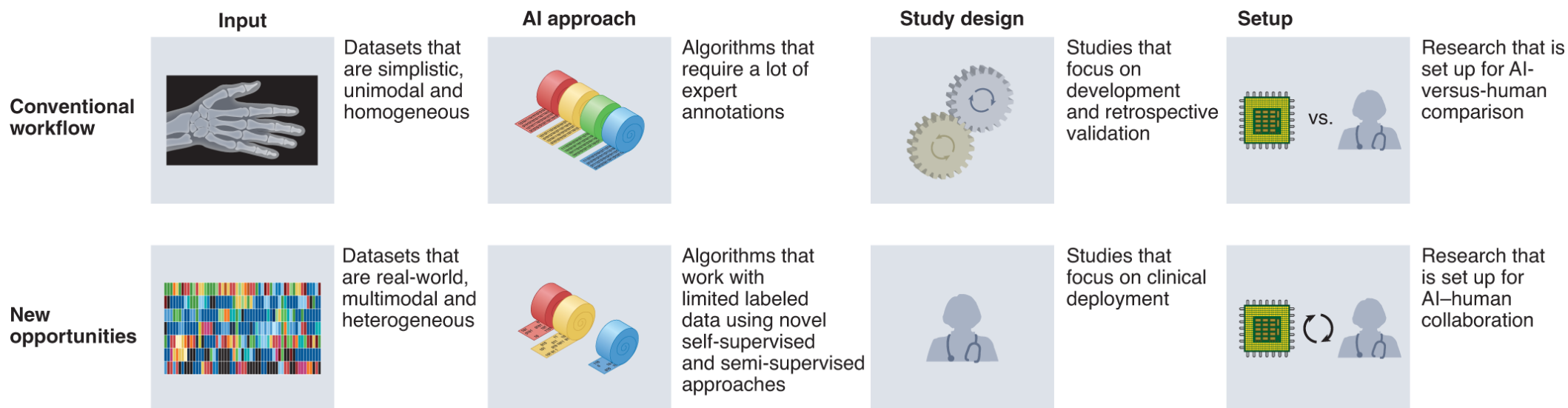


GENERATIVE AI



DIVERSE USE
CASES

CONVENTIONAL VS. NEW



Opportunities for the development of AI algorithms.

PRINCIPLES FOR RESPONSIBLE DEVELOPMENT

Principle	Questions
1. Alleviate healthcare disparities	• What health disparities are reported for the present AI application? • How can the AI tool be designed to be accessible to and improve outcomes for the disadvantaged population? • What clinical interventions are needed to realize the benefit, and are these accessible? • How can data collection be supported in underserved communities for tool retraining over time?
2. Report clinically meaningful outcomes	• How is clinical benefit defined in this domain? • What is the present threshold for the clinical benefit of existing tools, and how can the AI tool improve upon this threshold?
3. Reduce overdiagnosis and overtreatment	• What disease state is an overdiagnosis? • For every case of overdiagnosis, what are the downstream costs to the patient and healthcare system? • How can this AI application reduce the number of overdiagnoses compared to existing approaches?
4. Have high healthcare value	• Is this AI tool addressing a high-priority healthcare need? • What would be the cost to the healthcare system in implementation, maintenance, and update? • What would be the cost to the patient who does and does not benefit from this tool? • Does this tool have high healthcare value, and if not, how can it be improved?
5. Incorporate biography	• What biographical data can be collected or carefully coded for the intended population? • How do these factors vary in the intended population? • How can these factors be included when developing AI tools?
6. Be easily tailored to the local population	• Can the training features be easily collected in different settings? • Are these features reliable for training across different populations? • Will the AI/ML workflow be made open-access?
7. Promote a learning healthcare system	• How will this AI application be evaluated over time, and at what intervals? • What are acceptable thresholds for performance? • How will the evaluation results contribute to continuous improvement?
8. Facilitate shared decision-making	• Have AI explainability tools been explored and utilized? • Do clinicians and patients find the explainability results helpful? • Have simpler, explainable algorithms been tried and compared to 'black-box' algorithms to determine if a simpler model performs just as well? • How can patient values be easily integrated into the use of the AI tool?

Questions that can be used when considering each principle in the AI development process.

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ANALYZE CASE STUDIES IN REAL-WORLD EXAMPLES

PHASED RESEARCH FRAMEWORK OF A-GPS

AMIA AI Evaluation Showcase



Stage I paper

A Technical Performance Study and Proposed Systematic and Comprehensive Evaluation of an ML-based CDS Solution for Pediatric Asthma

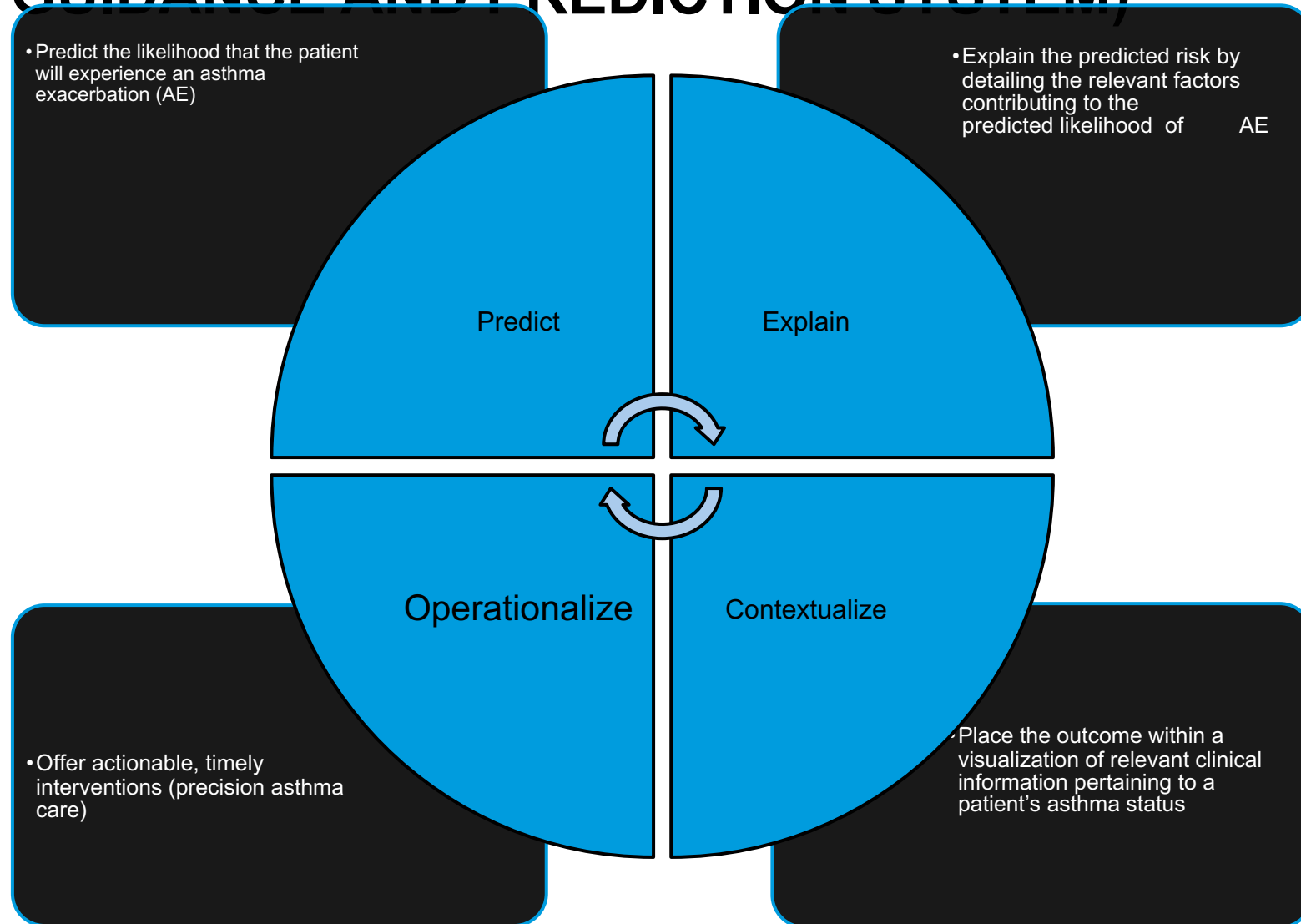


Stage II paper

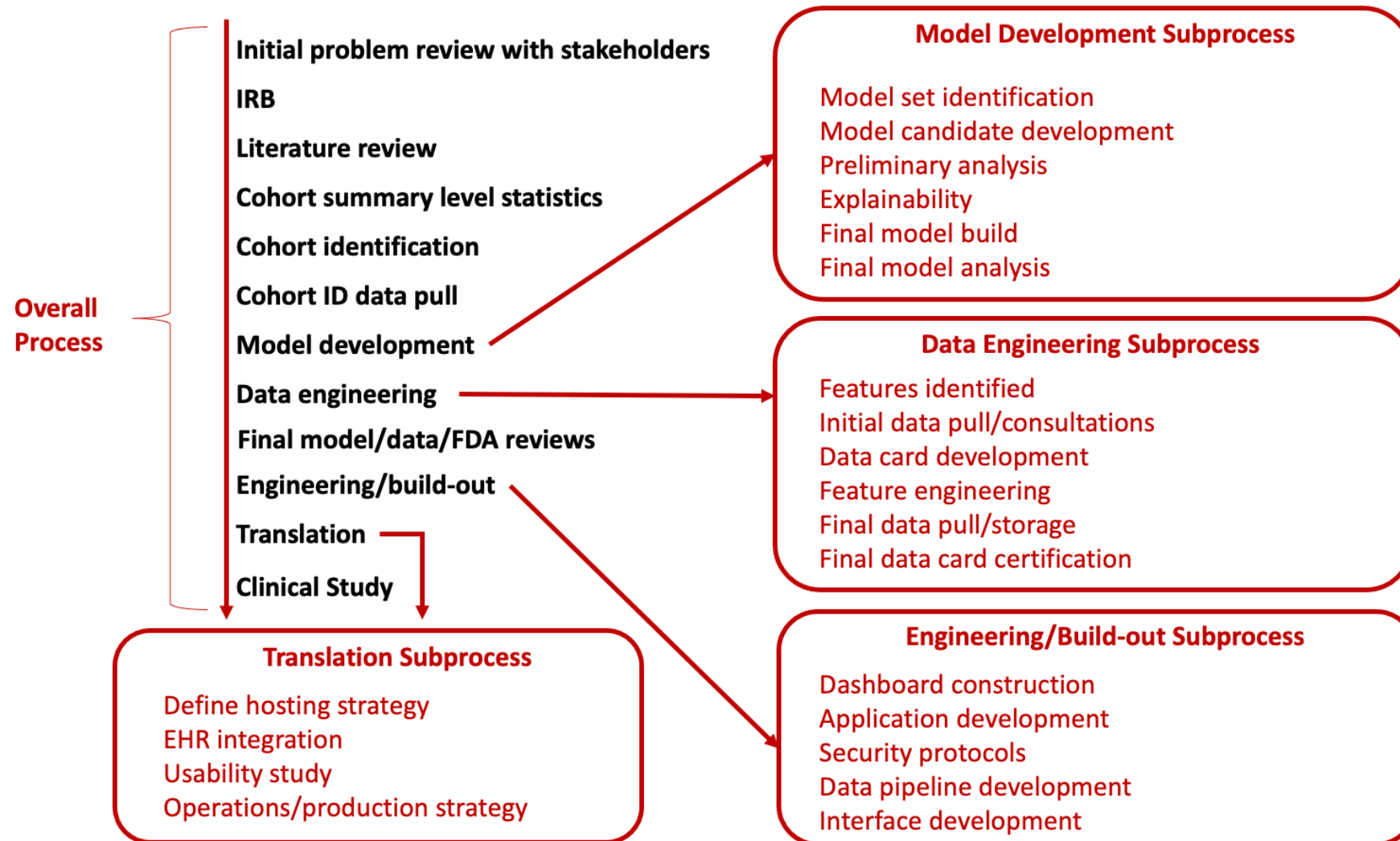
Stage III paper

Phase 0: Discovery and Intervention	Phase 1: Technical performance and safety	Phase 2: Efficacy and side effects	Phase 3: Therapeutic Efficacy	Phase 4: Safety and Effectiveness
 - User-centered design work and user requirements specification through stakeholder interviews - Evaluation of user requirements against NASSS framework - Map model outcomes to future prospective study or clinical trial	 - IRB and FDA submission - Algorithm model explainability - UX design research through stakeholder shadowing and interviews - Prototype design - Deployment feasibility analysis - Data quality evaluation	 - Iterative interface design with stakeholder focus groups - Pre-pilot model performance and validation assessment - Model implementation evaluation and validation - Business plan validation and deployment planning	 - Prospective study - Clinical validation and model documentation	 - Post-deployment surveillance and maintenance risk - Quality monitoring and audit - Model maintenance - Corrective and preventative action model documentation
Prepare	Develop	Validate	Deploy	Maintain
Model Documentation Framework (1)				

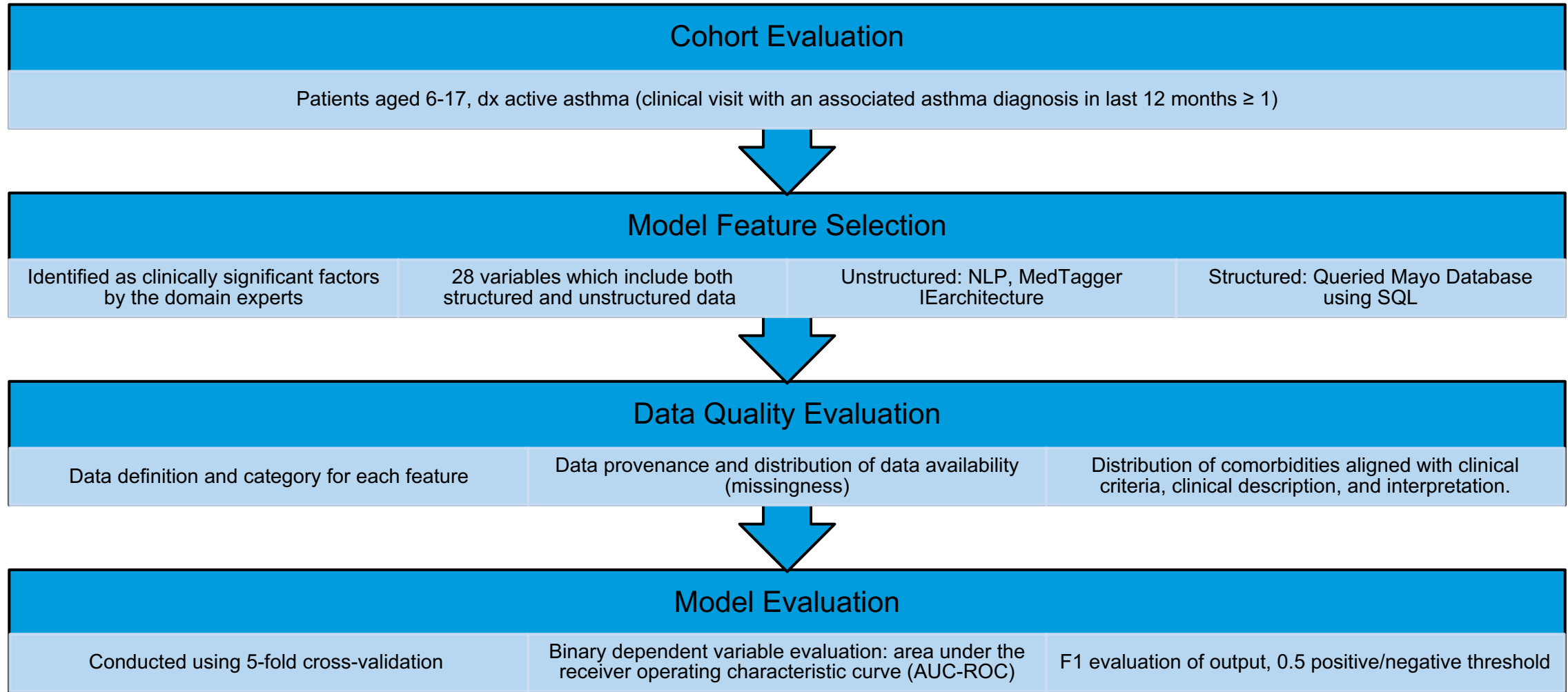
AIM OF A-GPS (ASTHMA GUIDANCE AND PREDICTION SYSTEM)



METHODOLOGY



MODEL DEVELOPMENT



DATA CARD

Metadata Prototype

Metadata Prototype				
Overall capabilities: FAIR principles, metadata completeness reports, traceability, usage metrics, version control, data governance, comment section for each card aspect, suggested metadata elements based on other projects, executable, searchable and discoverable, actionable and reproducible, bottom-up crowd sourcing				
	Project Card	Data Card	Model Card	
Scope	The Project Card contains the project description, inclusion/exclusion criteria for project scope, and includes suggestions for inclusion/exclusion criteria from an electronic phenotype database (PheKB). The Project Card sets the objective and scope of the project, describing the overall clinical intent, limitations, clinical departments involved, etc.	The Data Card contains a feature description, distributions over time, frequencies within the study cohort and population subsets, associated code sets, data availability and completeness, and data source. The Data Cards will be model-agnostic in order for Data Cards to be reusable across projects.	The Model Card includes the AI model description, inputs and outputs of the model, model architecture, limitations, trade-offs, missingness handling technique, and model performance. The model is dependent on the data features used in the model, therefore there are many Data Cards to one Model Card. The Model Card operates on the defined population.	
	clinicaltrials.gov	Mayo Clinic Data Card Template	Model Cards for Model Reporting	
	Design description: The Clinical Trials website was created to "provide patients, family members, health care professionals, researchers, and the public easy access to information on publicly and privately supported clinical studies." Each study is maintained and updated appropriately to ensure relevancy and transparently communicate the description of a clinical trial. The NIH and FDA collaborated to create this registry to "test the effectiveness of experimental drugs for serious or life-threatening diseases or condition" after the FDAMA. The components included in each study follows a defined set of standard reporting protocols that compiles a summary about the study. Design components: Study Description Eligibility Criteria Study Design Contacts and Locations Groups and Cohorts More Information Outcome Measures	Template description: A Data Card represents example metadata that would be needed for documenting the features of a machine learning model. Data Cards support transparency, accountability, and explainability the reporting of model features by standardizing the documentation of feature descriptions. Data Card components: Project Information Structured Data (Feature Name...) Unstructured Data (Feature Name...)	Journal description: Model Cards are a documentation framework accompanying ML models that acts as a transparency tool for model reporting by providing descriptions of functionality and performance metrics while also including metrics that capture bias, fairness and inclusion considerations. Model Cards aim to provide a concise, holistic view of ML models by explaining what it does, its intended audience, who maintains it, model construction and architecture, and information on the training data, detailing a model's performance metrics as well as limitations and risk mitigations. Model Card components: Model Details Intended Use Factors and Metrics Evaluation Data	Training Data Quantitative Analyses Ethical Considerations Caveats and Recommendations

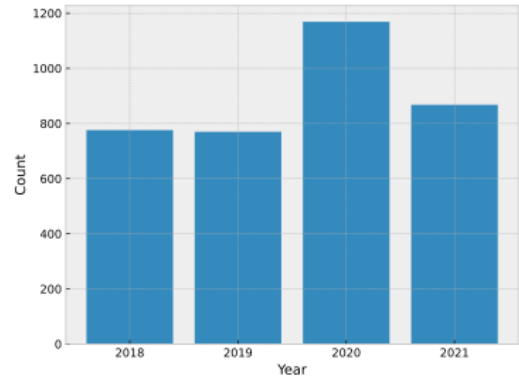
Data Card Example

❖ Sample Query (UDP)

```
SELECT
  P.PATIENT_CLINIC_NUMBER,
  FS.PATIENT_AGE_AT_EVENT,
  FS.SMARTFORM_ORIGINAL_DTM,
  FS.SMARTFORM_UPDATE_DTM,
  FS.SMARTFORM_CONCEPT_NAME,
  FS.SMARTFORM_ANSWER
FROM EDTW.HFACT_SMARTFORM FS
LEFT JOIN EDTW.DIM_PATIENT P
ON FS.PATIENT_DK = P.PATIENT_DK
LEFT JOIN EDTW.DIM_SMARTFORM_CONCEPT DSC
ON FS.SMARTFORM_CONCEPT_DK = DSC.SMARTFORM_CONCEPT_DK
WHERE P.PATIENT_CLINIC_NUMBER IN (XYZ)
AND DSC.SOURCE_SYSTEM_KEY Like 'AAP#%'
AND FS.PATIENT_AGE_AT_EVENT < 18
Order By FS.PATIENT_DK, FS.SMARTFORM_UPDATE_DTM,
FS.SMARTFORM_ANSWER_HEADER, FS.SMARTFORM_ANSWER_SEQUENCE
WITH UR;
```

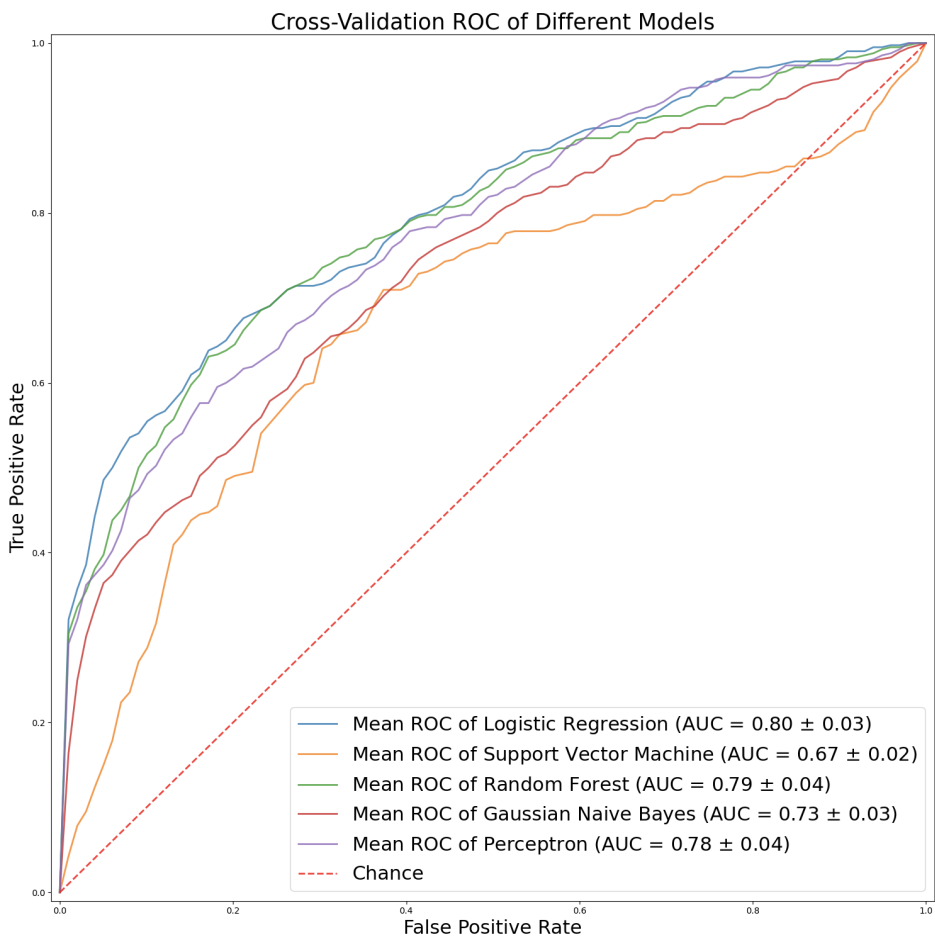
❖ Data Source
The above query has been collected from the UDP database.

❖ Distribution of data availability
➤ Study cohort: Pediatric asthma registry as of 6/10/2021 (MCR) (n=1889)
➤ Yearly Distribution for AAP Update:



RESULTS

Comparison of AUC score of various models



Final parameters chosen for the logistic regression model were: (C=0.23357214690901212, max_iter=10000, penalty='l1', solver='liblinear'). All other parameters of logistic regression were applied with default values defined in Scikit-learn package

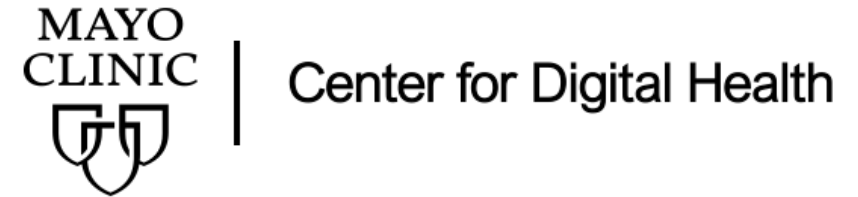
Prediction Task
Predict asthma exacerbation (AE)
within the next 12 months

Logistic Regression model output

Threshold	Mean F1
0.1	0.421527
0.2	0.522789
0.3	0.531575
0.4	0.484955
0.5	0.456748
0.6	0.430991
0.7	0.4161
0.8	0.388956
0.9	0.339465

MULTIDISCIPLINARY TEAM

- United objective to provide solutions that are:
 1. People centric
 2. Value added
 3. Evidence-based



Multidisciplinary Team to Facilitate Translation				
Translational Informatics Analysts	Clinicians	Data Scientists	Ethicists	MLOps
Product Owner	Operational Staff	Patients	Quality Management Team	UX Researchers and Designers

TRANSLATION AND DEPLOYMENT OF A-GPS

Translation and Deployment of AI Models into Production

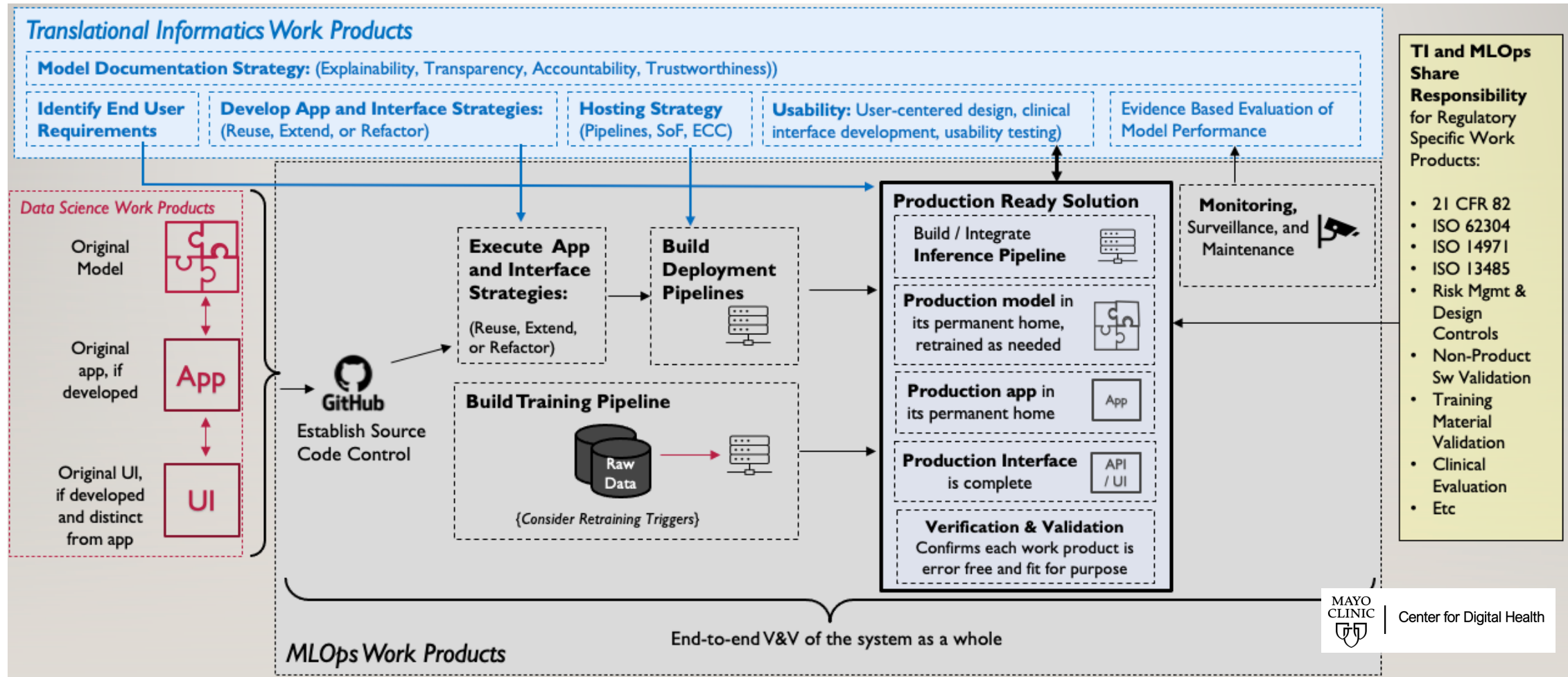


Diagram Credit: John Skiffington, MGR SaMD Software Engineering

A Proposed Model Documentation Framework to Facilitate Translation of AI Models in Healthcare

Tracey A. Brereton¹, Shauna M. Overgaard, Ph.D.¹, Kristelle A. Jose¹, Chung II. Wi, M.D.^{2,3}, Kevin J. Peterson, Ph.D., M.S.¹, Amin Nikakhtar, Ph.D., M.S.¹, Young J. Juhn, M.D.^{2,3}

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³Department of Pediatric and Adolescent Medicine, Mayo Clinic, Rochester, Minnesota, United States of America

What might the attendee be able to do after being in your session?

Attendees will be equipped to apply a Model Documentation Checklist to guide AI/ML (Artificial Intelligence / Machine Learning)-based clinical decision support (CDS) tool documentation, encompassing multidisciplinary best practice reporting requirements to facilitate the translation of explainable and transparent AI/ML tools in healthcare. The Model Documentation Checklist is a collaborative tool that a) follows phases of product lifecycle and AI clinical evaluation, b) serves to iteratively develop a comprehensive AI model documentation deliverable, and c) rigorously addresses translation, implementation, and accountability gaps in oversight and standardization by leveraging reporting guidelines and published best practices.^{1,2,3-6}

Description of the Problem or Gap

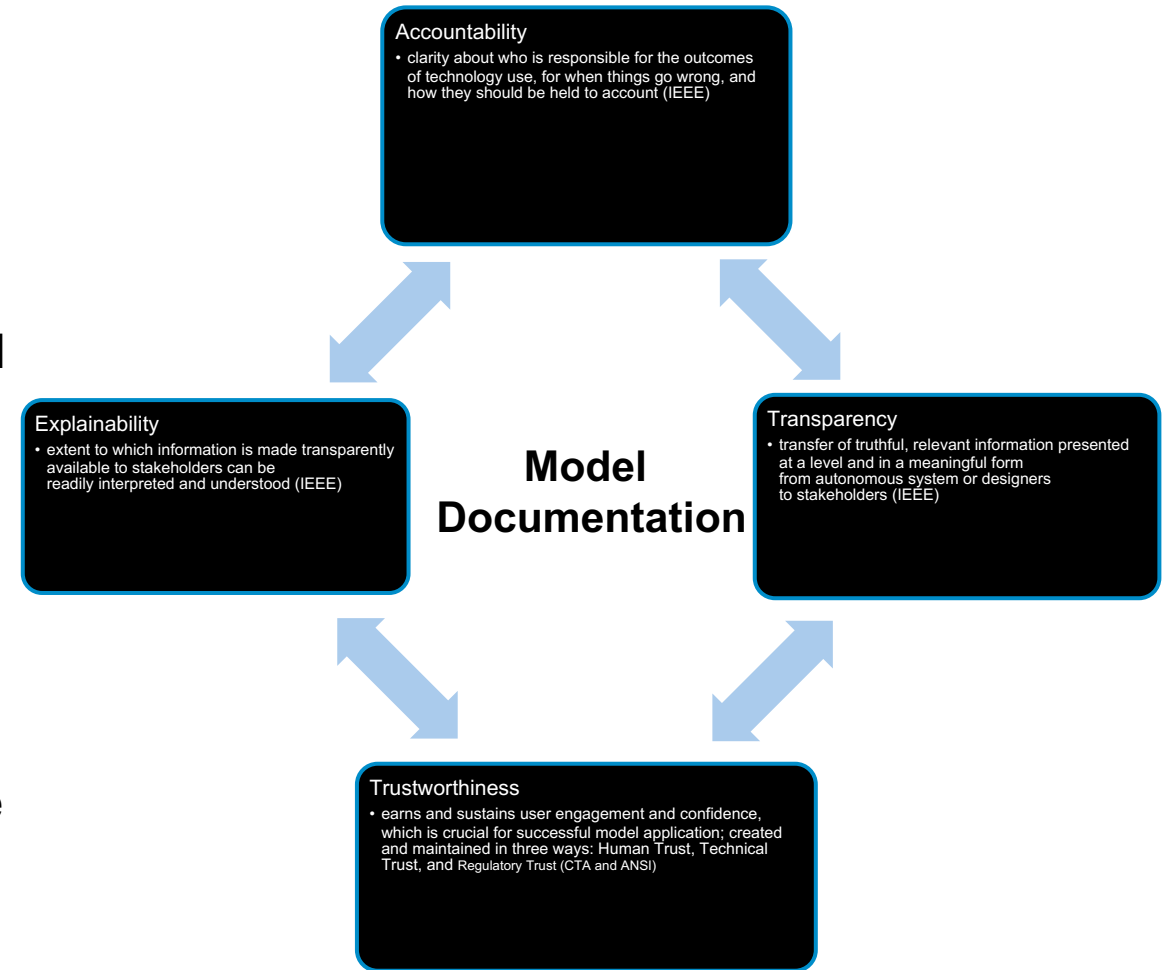
Literature surrounding the current state of AI/ML-based CDS documentation recognizes inconsistencies in standards and governance that do not effectively balance CDS tool intelligence with explainability, therefore creating gaps in translation, implementation, and accountability.² CDS tools integrated within clinical settings show evidence of accompanying documentation with levels of rigor and transparency regarded as insufficient by field experts and justifications involving fairness, long-term reliability, and usefulness as limited, leading to biased, untrustworthy, black-box tools.^{7,8} The lack of sufficient documentation and inability to leverage available resources to communicate functionality proves challenging for CDS tools to be explainable from technological, legal, medical, and patient perspectives and must be enhanced to effectively translate tools into clinical settings.⁹

Methods

To satisfy a real need in our organization's translational AI work, our interdisciplinary team sought to create a comprehensive model documentation framework, from which we derived an accompanying Model Documentation

AI/ML Documentation

- **Primary values driving model documentation framework: explainability, transparency, accountability, trustworthiness**
- Harmonizes AI development and ethical considerations
 - Complements and works in tandem with model development
 - Communicates model components and features
 - Addresses limitations and risks
 - Avoids complex "black-box" models
- Value and document transparency and accountability in model development to standardize explainable AI and gain trust



Sources: [ethical-dilemmas-in-ai-report.pdf](#) / [ANSI/CTA "The Use of Artificial Intelligence in Health Care: Trustworthiness"](#)

MODEL DOCUMENTATION FRAMEWORK

Model Documentation Checklist Items - Phase 1 - Prepare						
Patient Impact						
Completed / Reviewed	Assigned To	Index	Checklist Item / Description	Reporting Guideline	Section of Reporting Guideline	Referenced Item
		1-1	Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references to existing models	TRIPOD	Background and Objectives	3a
		1-2	Specify the objectives, including whether the study describes the development or validation of the model or both	TRIPOD	Background and Objectives	3b
		1-3	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including number and location of centers	TRIPOD	Participants (Methods)	5a
		1-4	Describe eligibility criteria for participants	TRIPOD	Participants (Methods)	5b
		1-5	Give details of treatments received, if relevant	TRIPOD	Participants (Methods)	5c
		1-6	Explain the intended use of the AI intervention in the context of the clinical pathway, including its purpose and its intended users (e.g. healthcare professionals, patients, public).	CONSORT-AI	Background and Objectives	2a
		1-7	Specify objectives, endpoints, and hypotheses	CONSORT-AI	Background and Objectives	2b
		1-8	Provide examples of out-of-scope use cases and note limitations of generalizability	Model Card	Intended Use	

[Tool Link](#)

ACKNOWLEDGMENTS

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- Stephanie Bernthal
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- Bryce Flor
- Joe Melnick
- Joshua Reed, PhD
- John Skiffington
- David Vidal, JD



Center for Digital Health

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